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(54) **ELECTRONIC DEVICE INCLUDING
STRUCTURE FOR SUPPORTING DISPLAY**

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CPC **G06F 1/1652** (2013.01)

(58) **Field of Classification Search**
CPC G06F 1/1652
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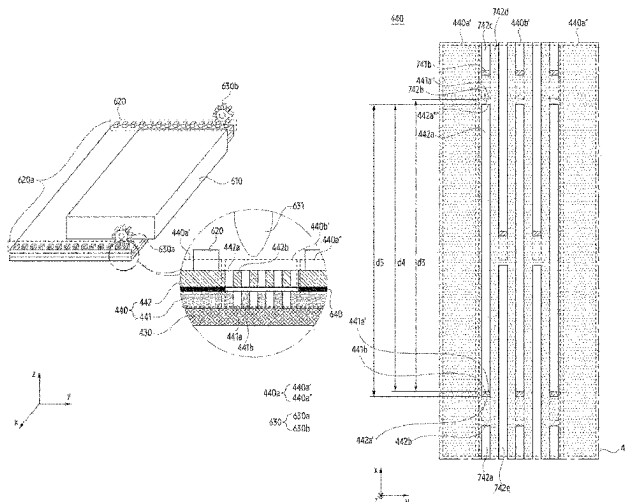
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(57) **ABSTRACT**

An electronic device according to an embodiment includes a first housing, a second housing coupled to the first housing so as to be slidable with respect to the first housing, a display including a planar region and a deformable region in contact with the planar region, and a support plate configured to supporting the display. The support plate includes rigid regions and a flexible region disposed between the rigid regions and including a plurality of openings. The support plate further includes a first layer including a plurality of first openings in the flexible region, and a second layer disposed on the first layer and including a plurality of second openings partially overlapping each of the plurality of first openings in the flexible region. Other embodiments are also possible.

20 Claims, 26 Drawing Sheets



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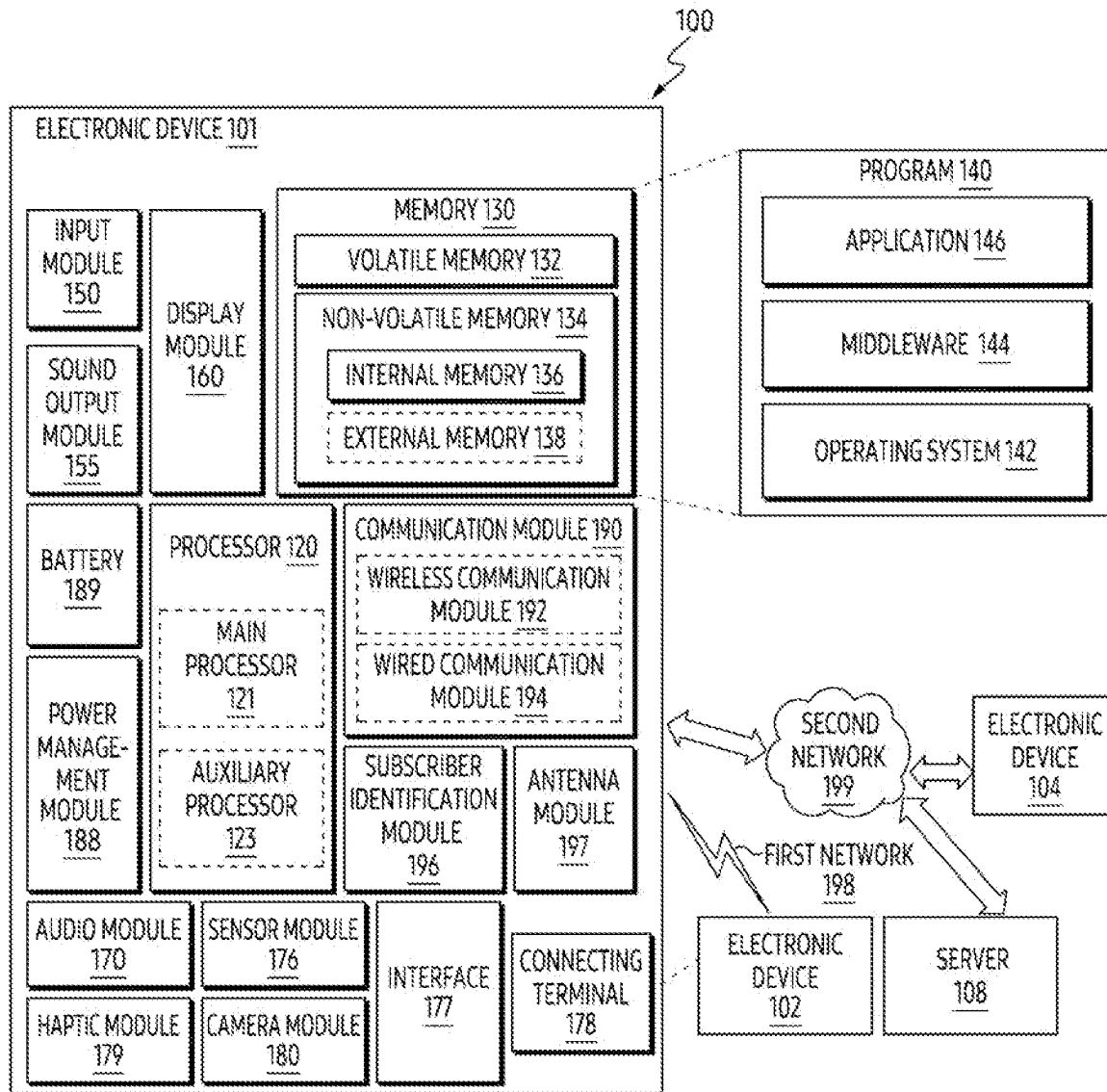


FIG. 1

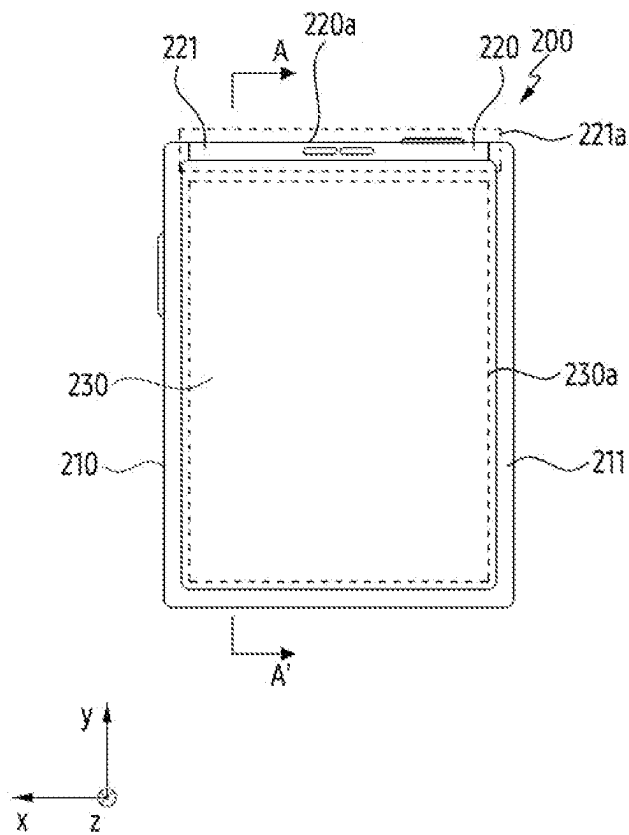


FIG. 2A

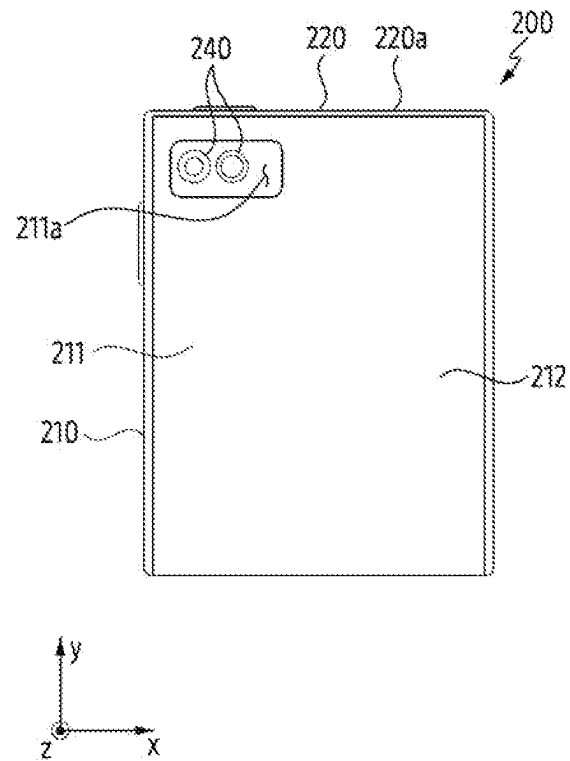


FIG. 2B

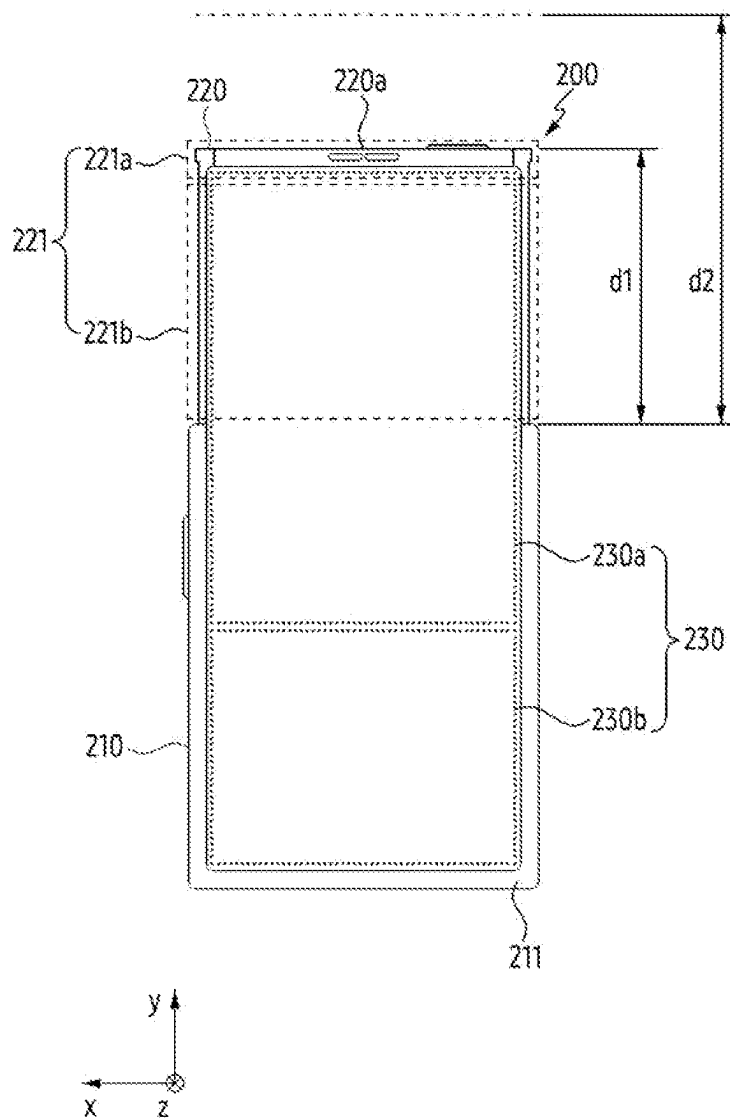


FIG. 2C

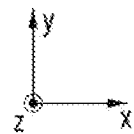
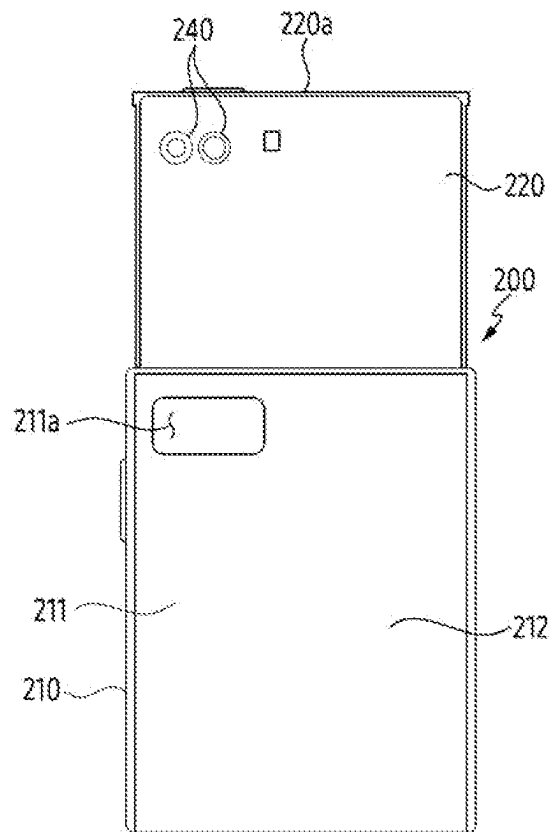
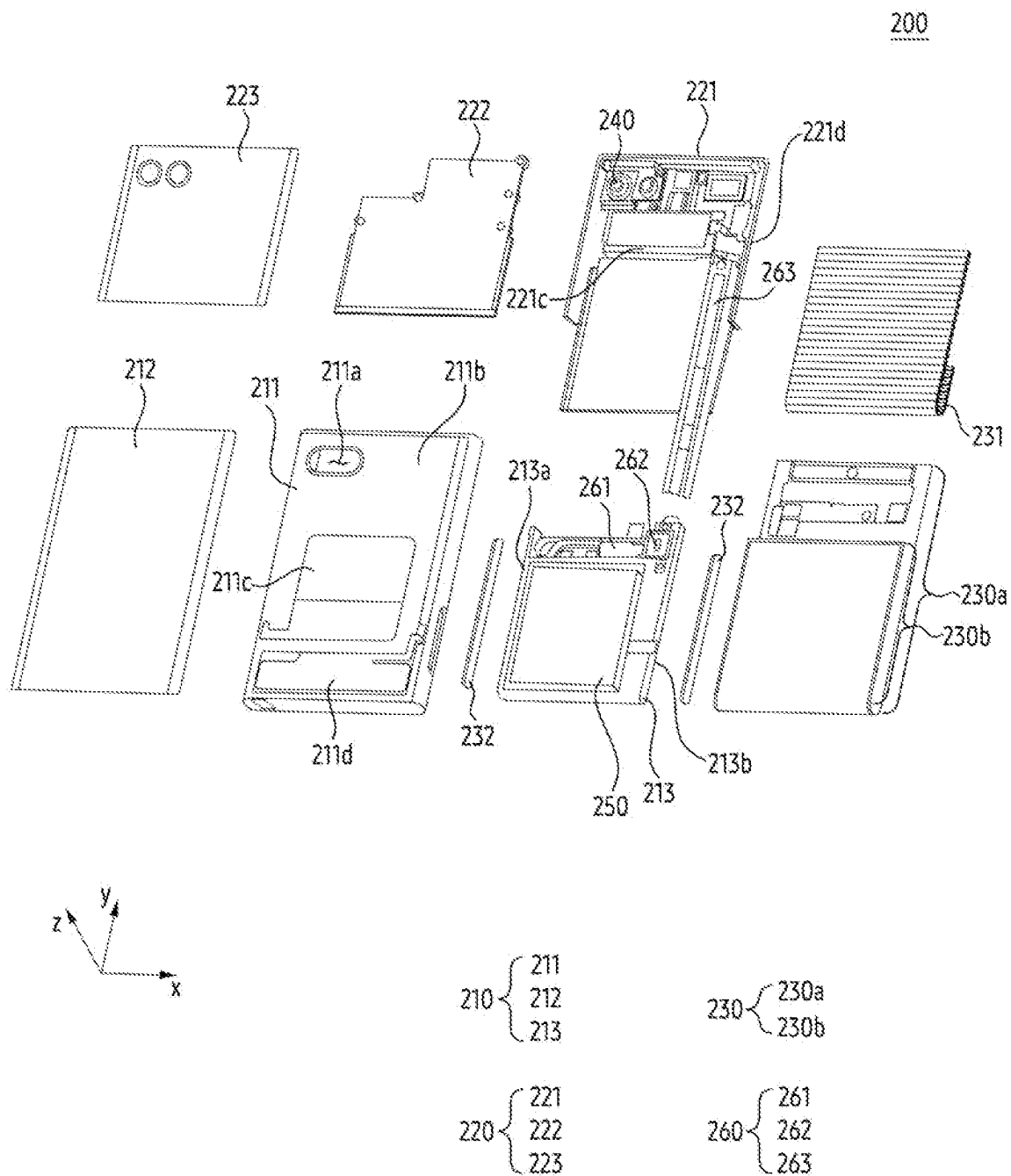


FIG. 2D



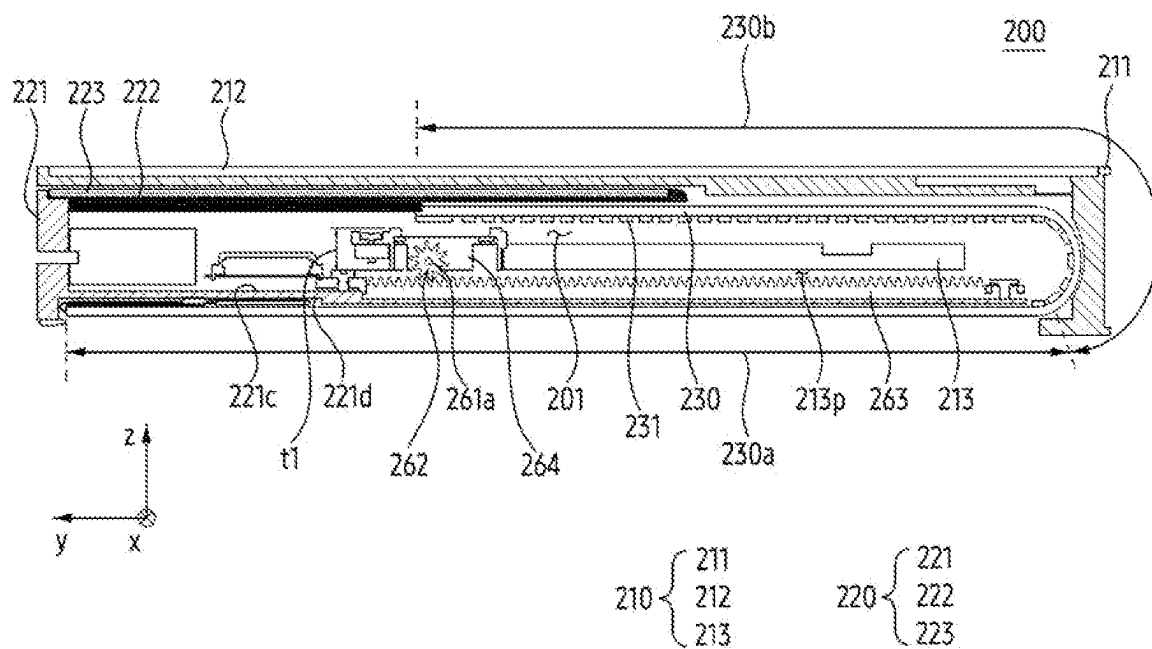


FIG. 3B

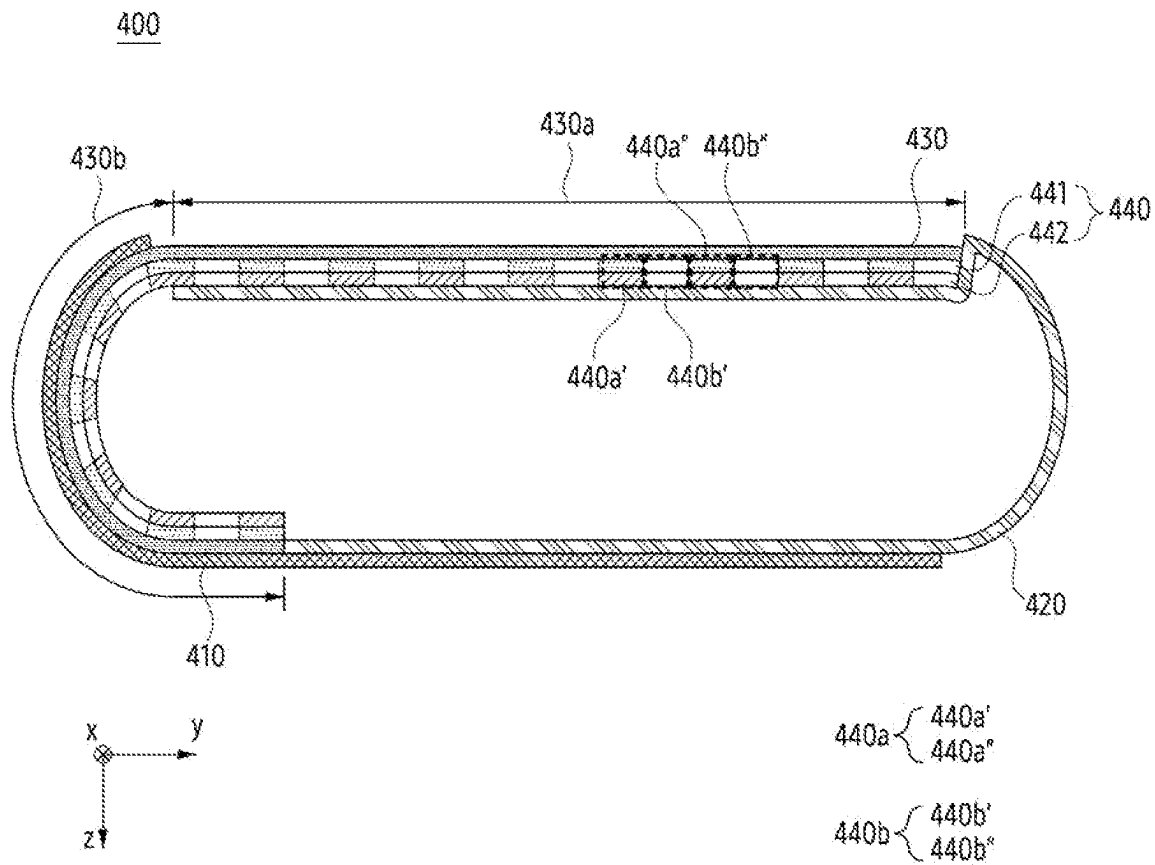


FIG. 4A

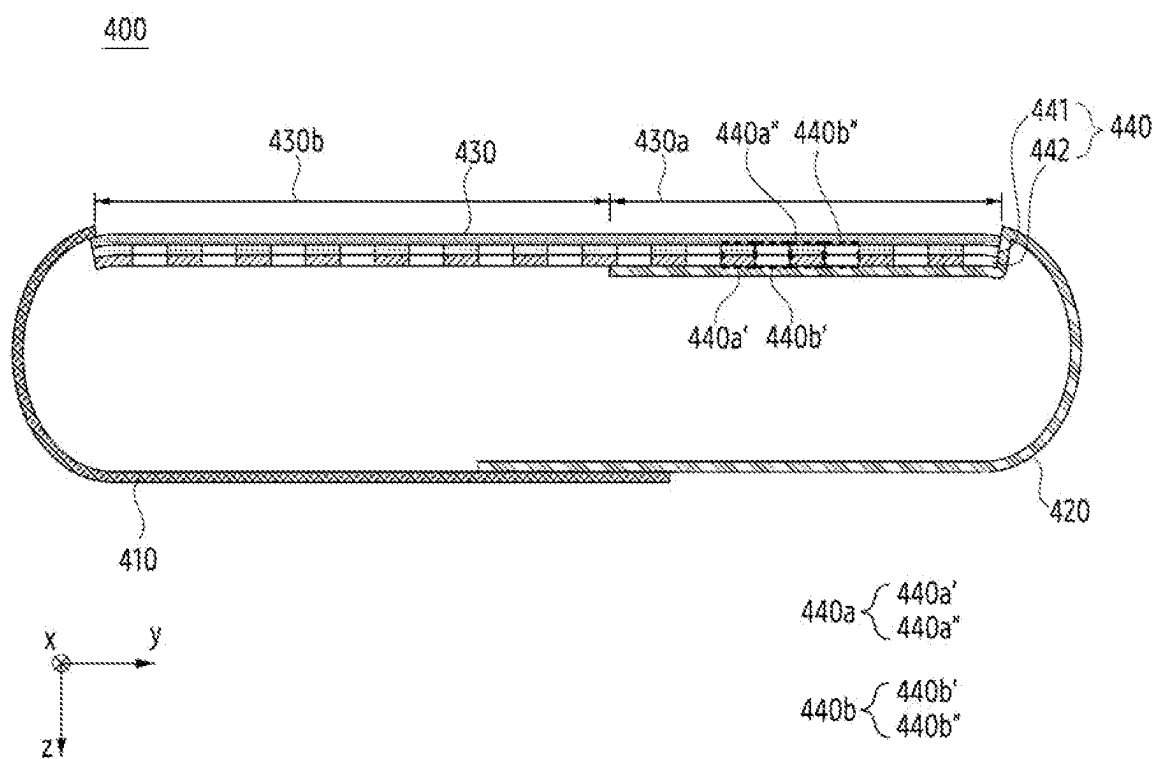


FIG. 4B

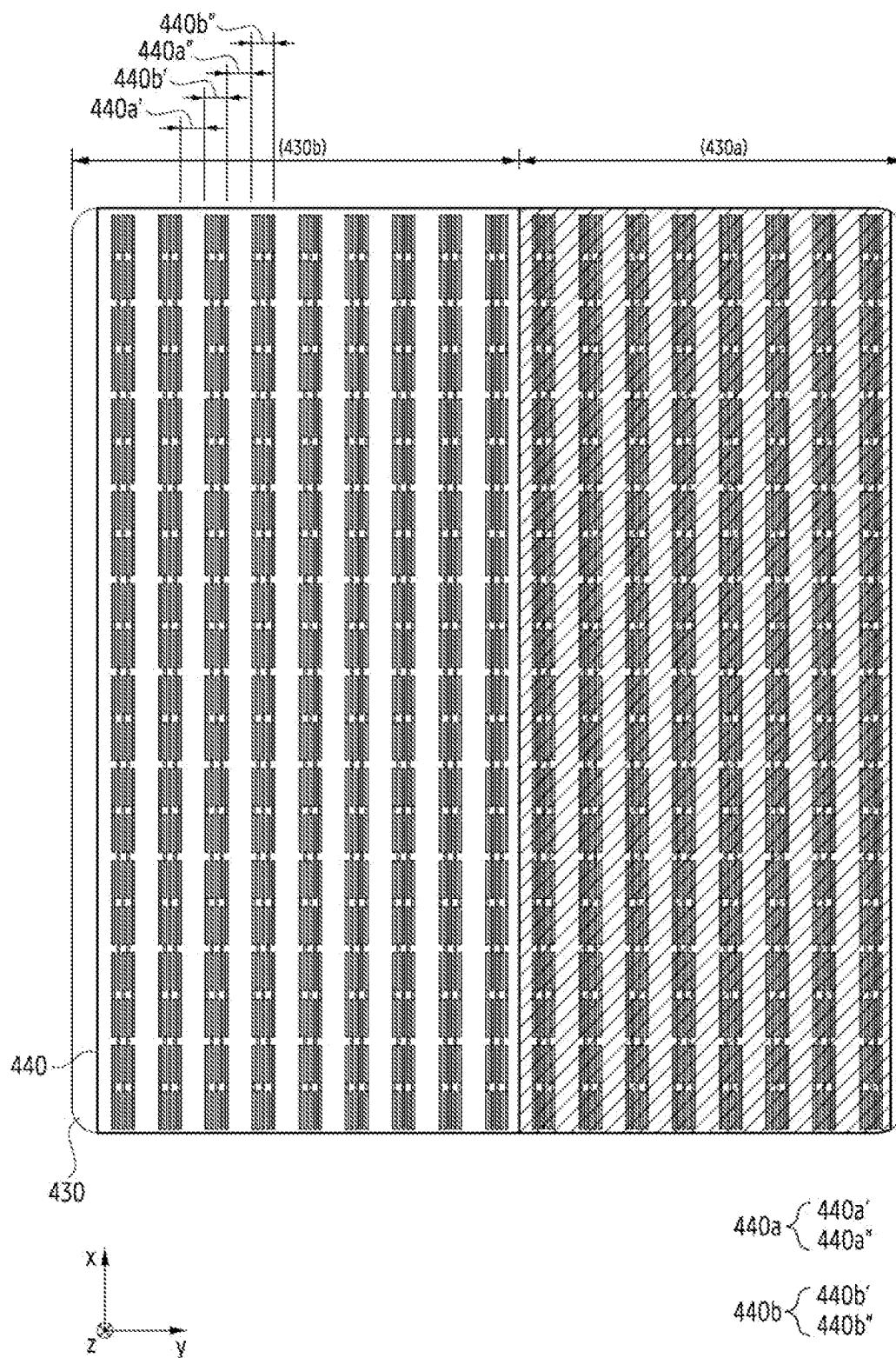


FIG. 5A

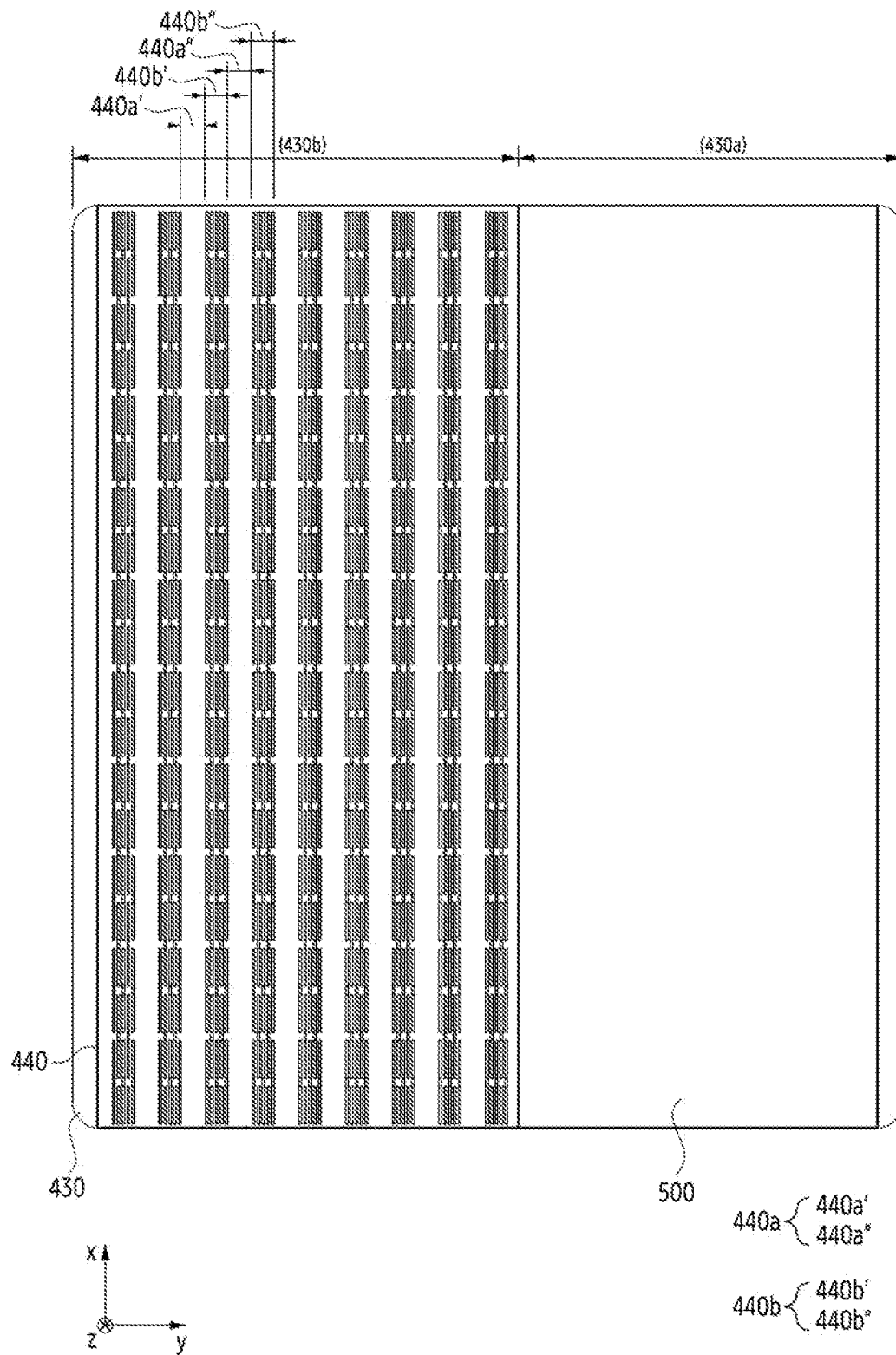


FIG. 5B

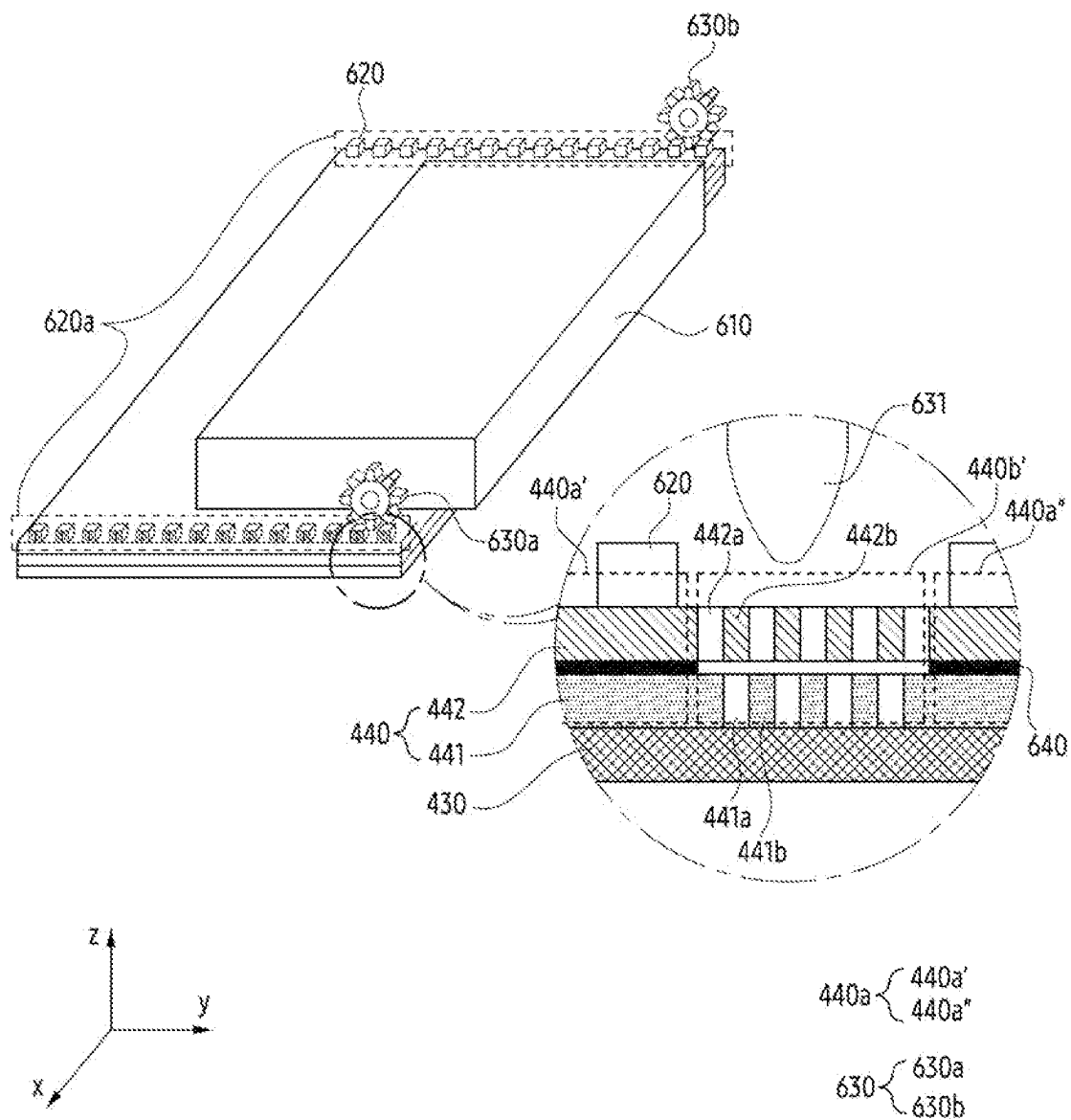


FIG. 6A

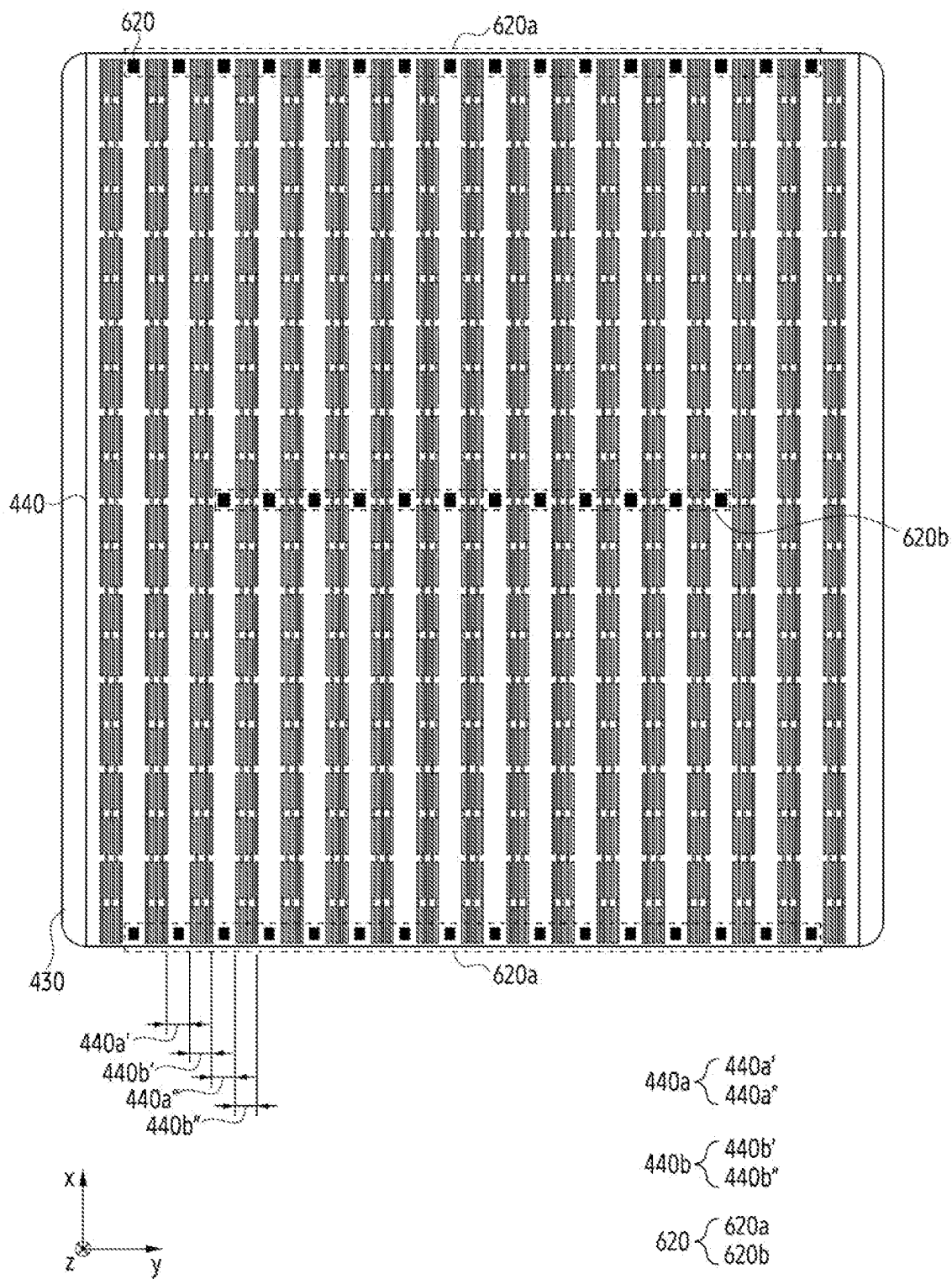


FIG. 6B

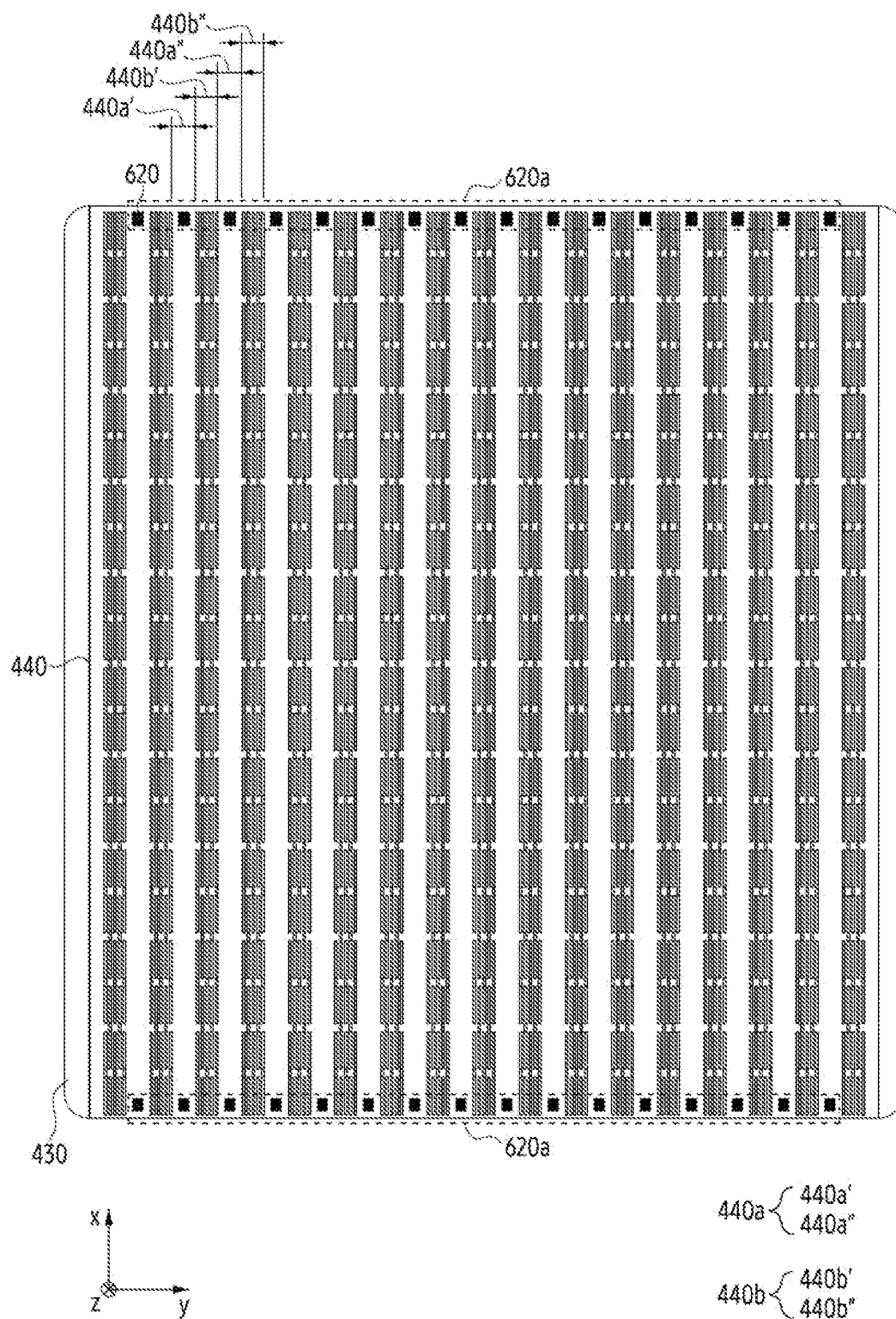


FIG. 6C

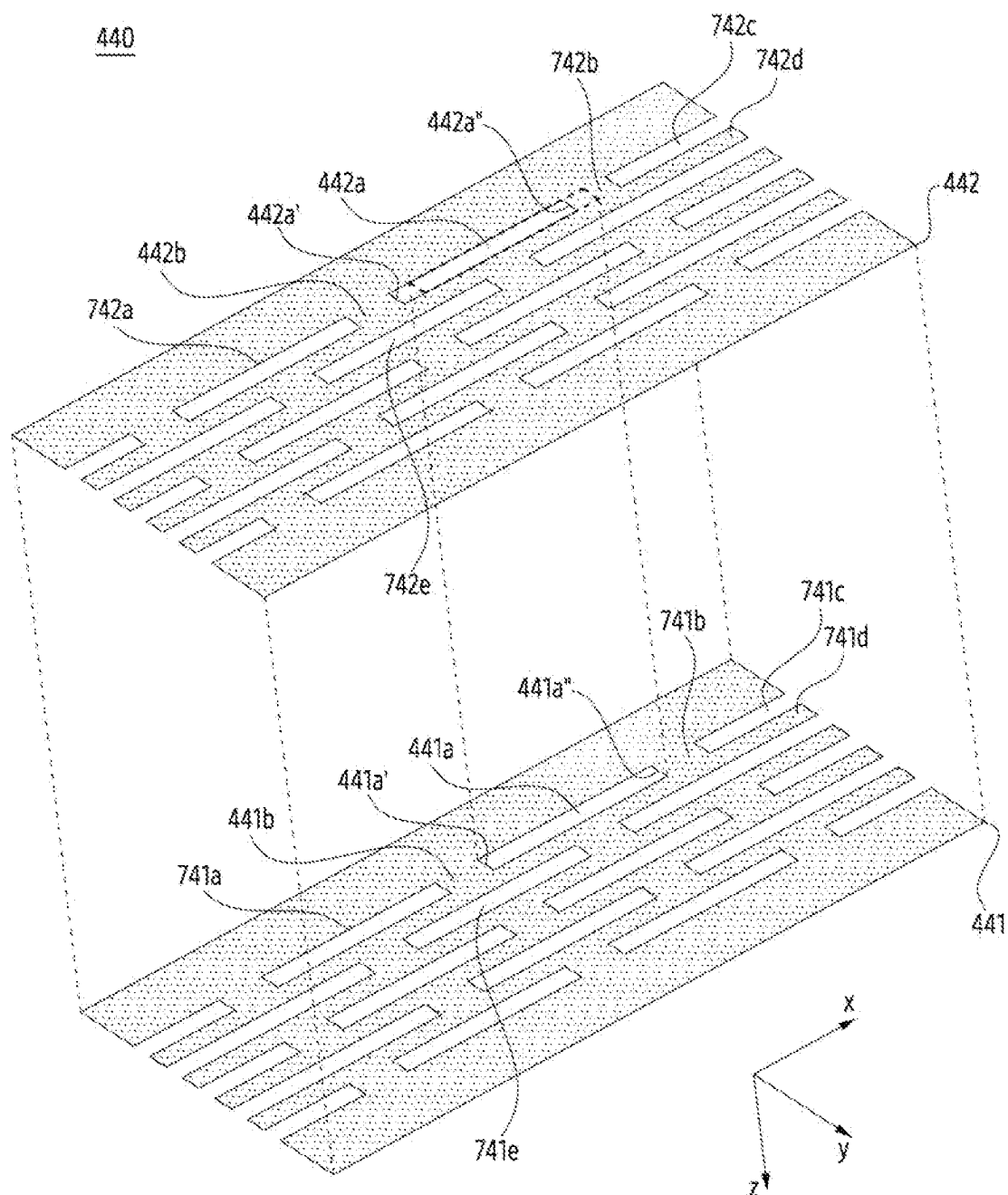


FIG. 7A

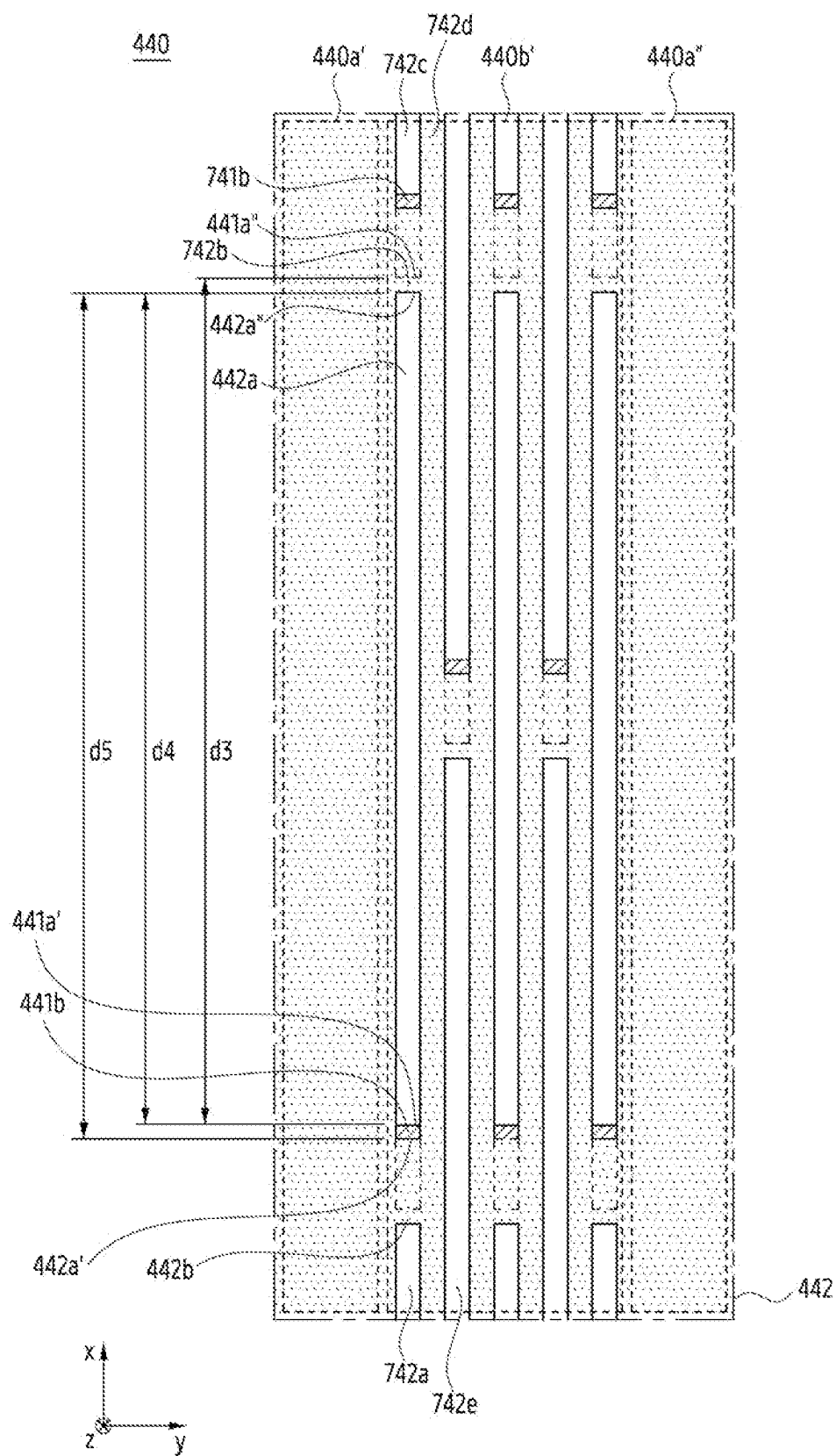


FIG. 7B

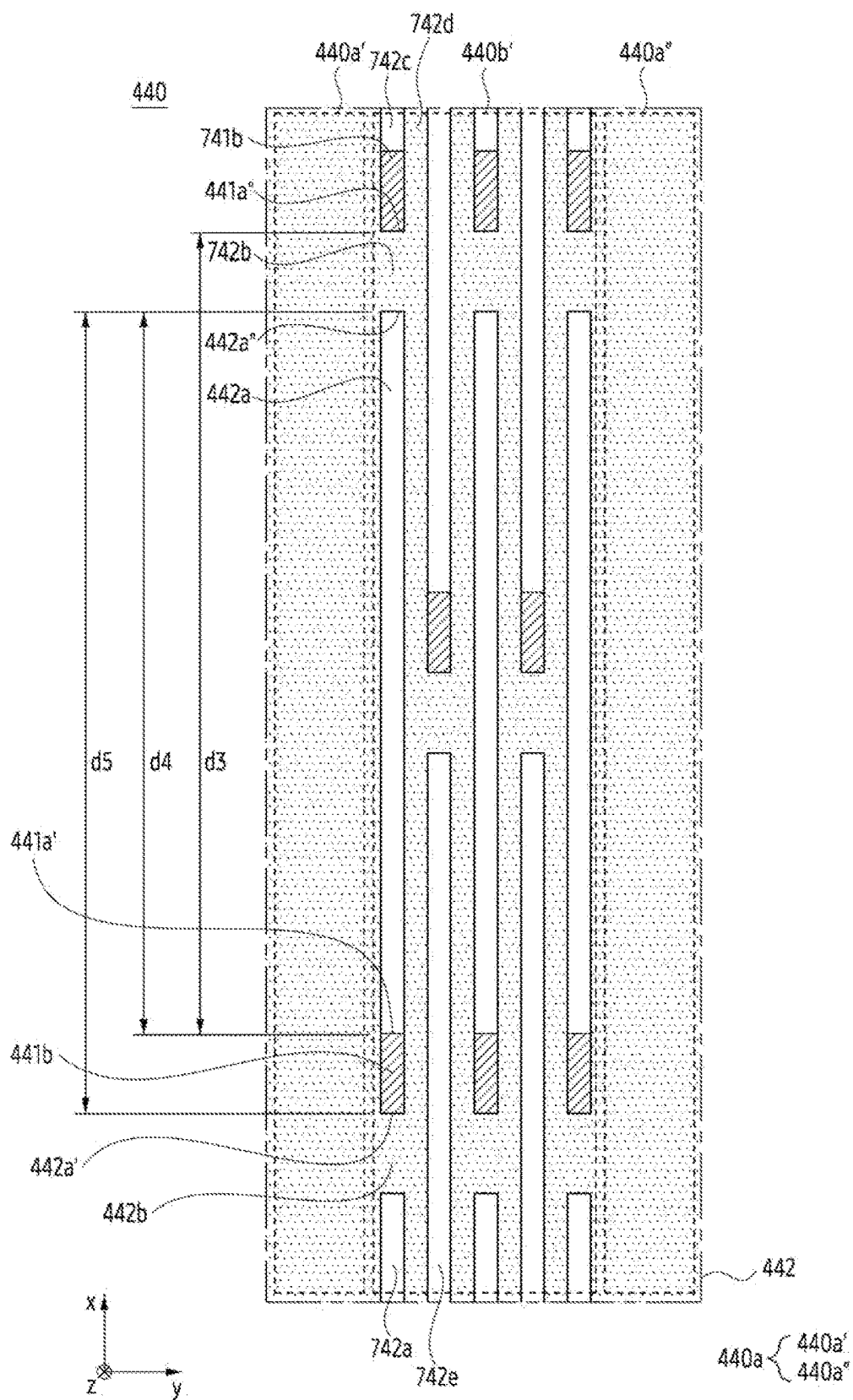


FIG. 7C

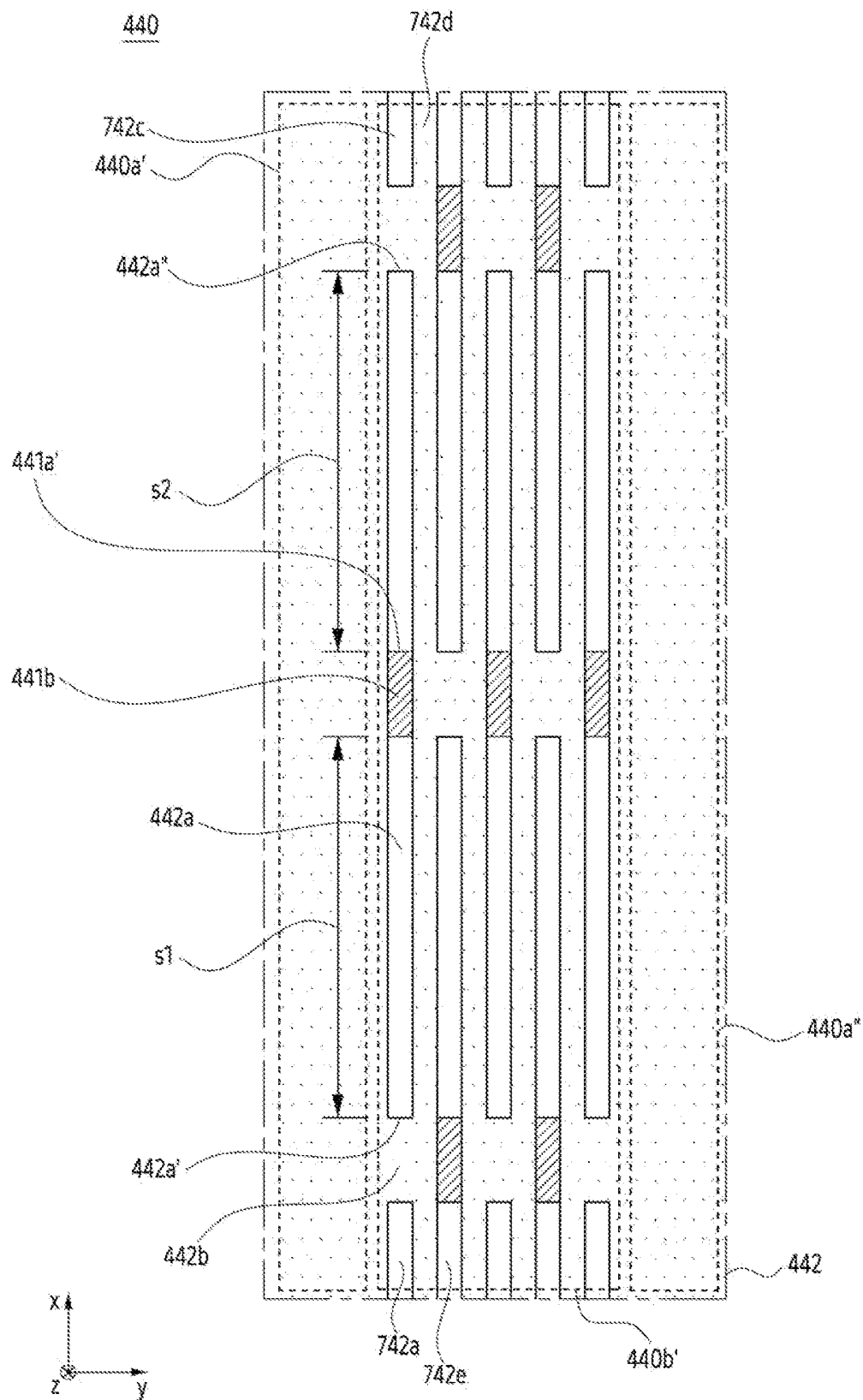


FIG. 7D

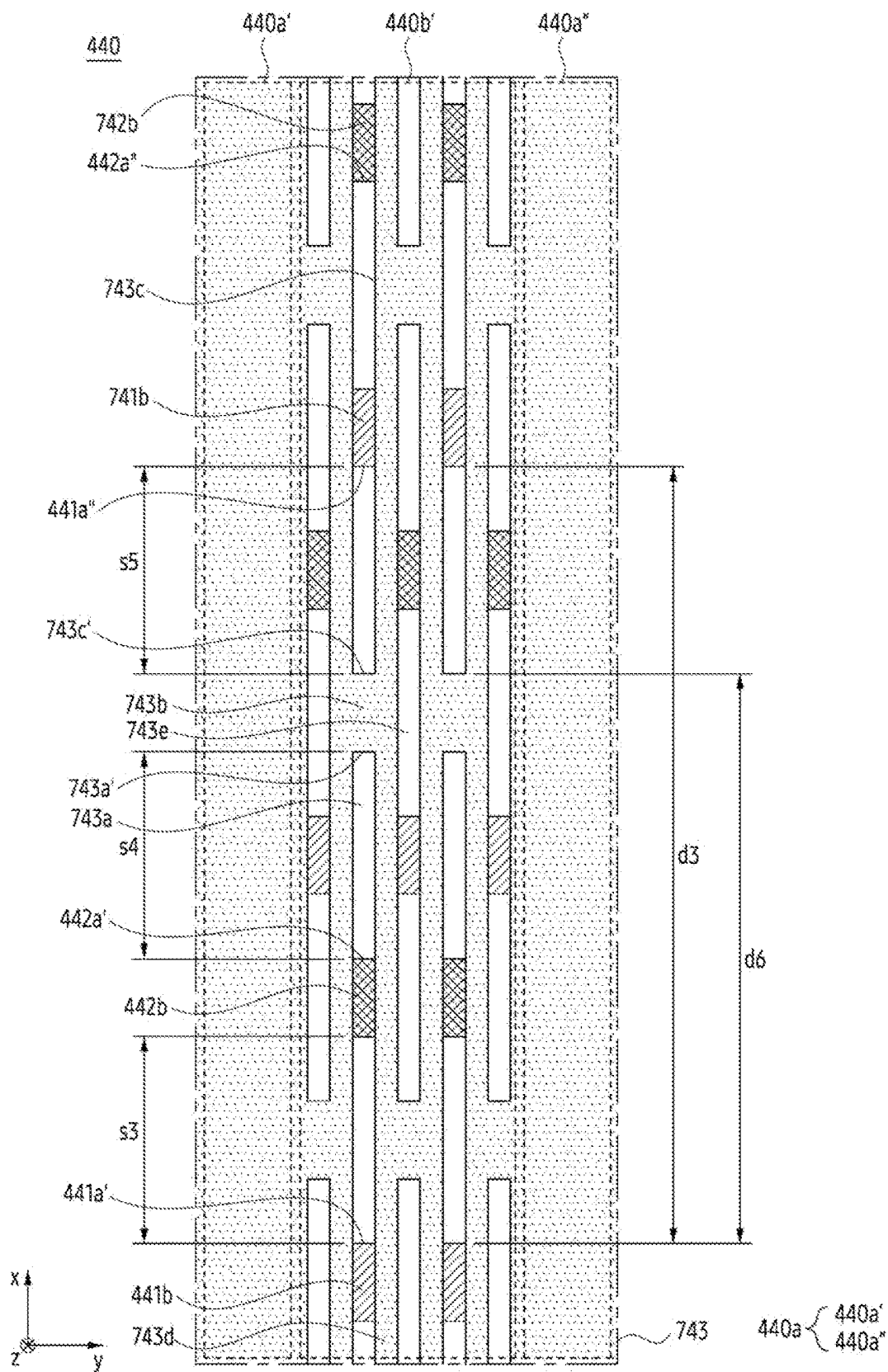


FIG. 7E

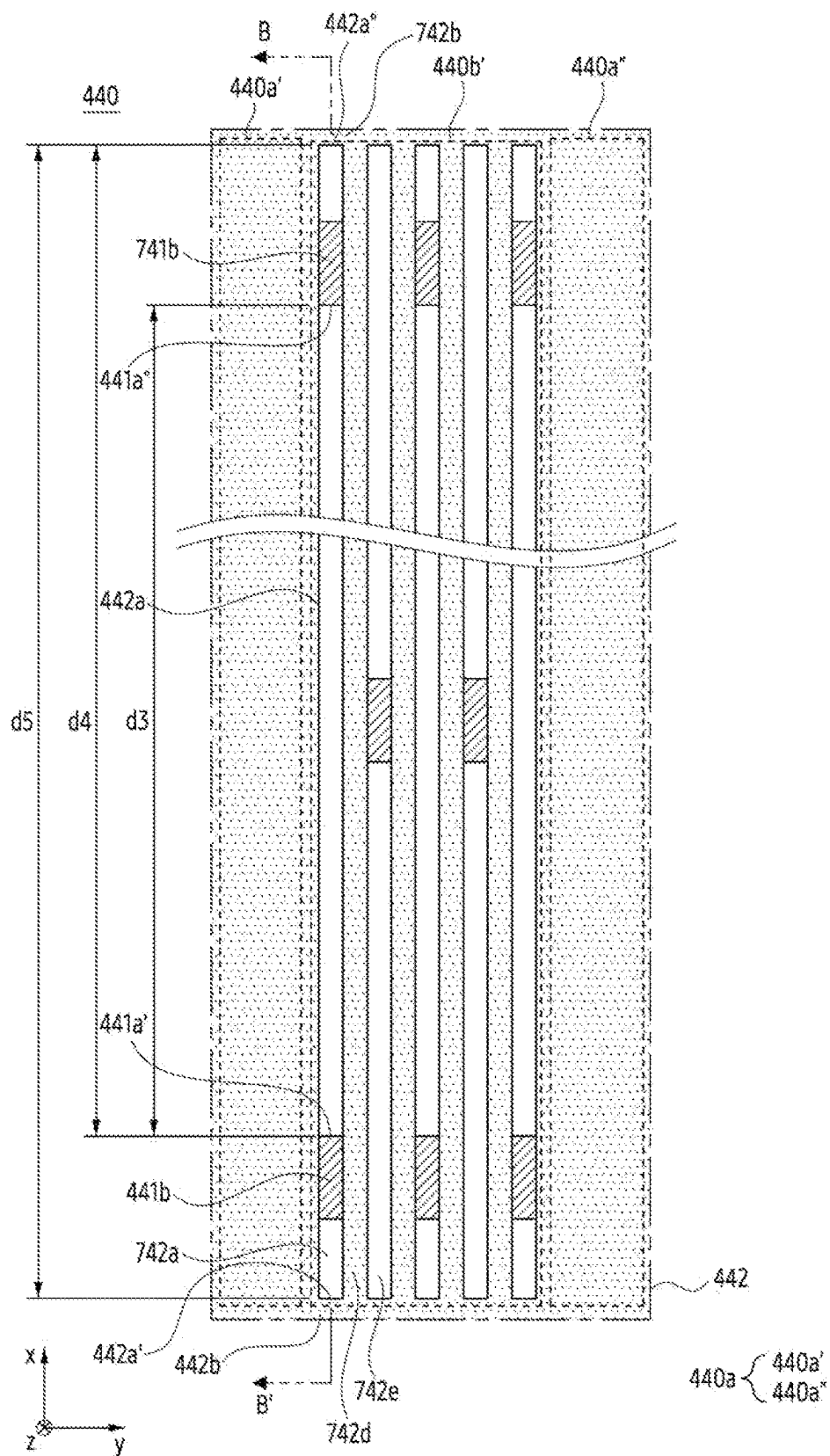


FIG. 8A

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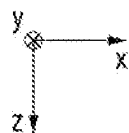
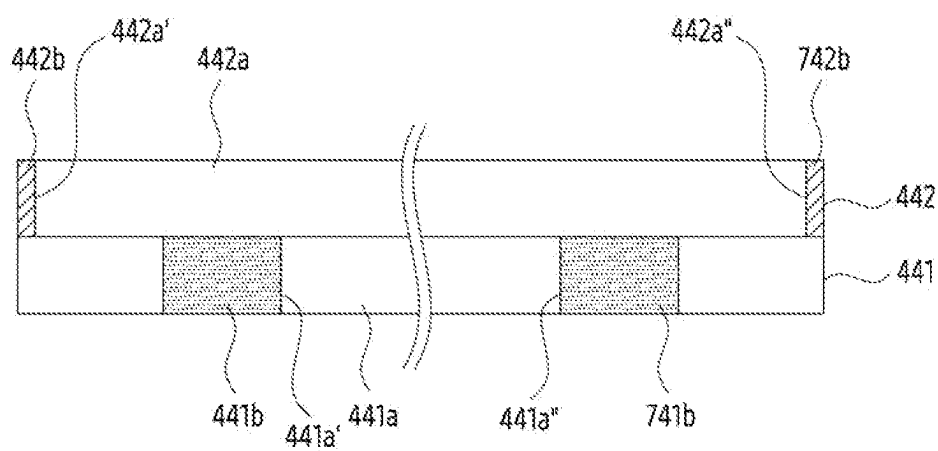


FIG. 8B

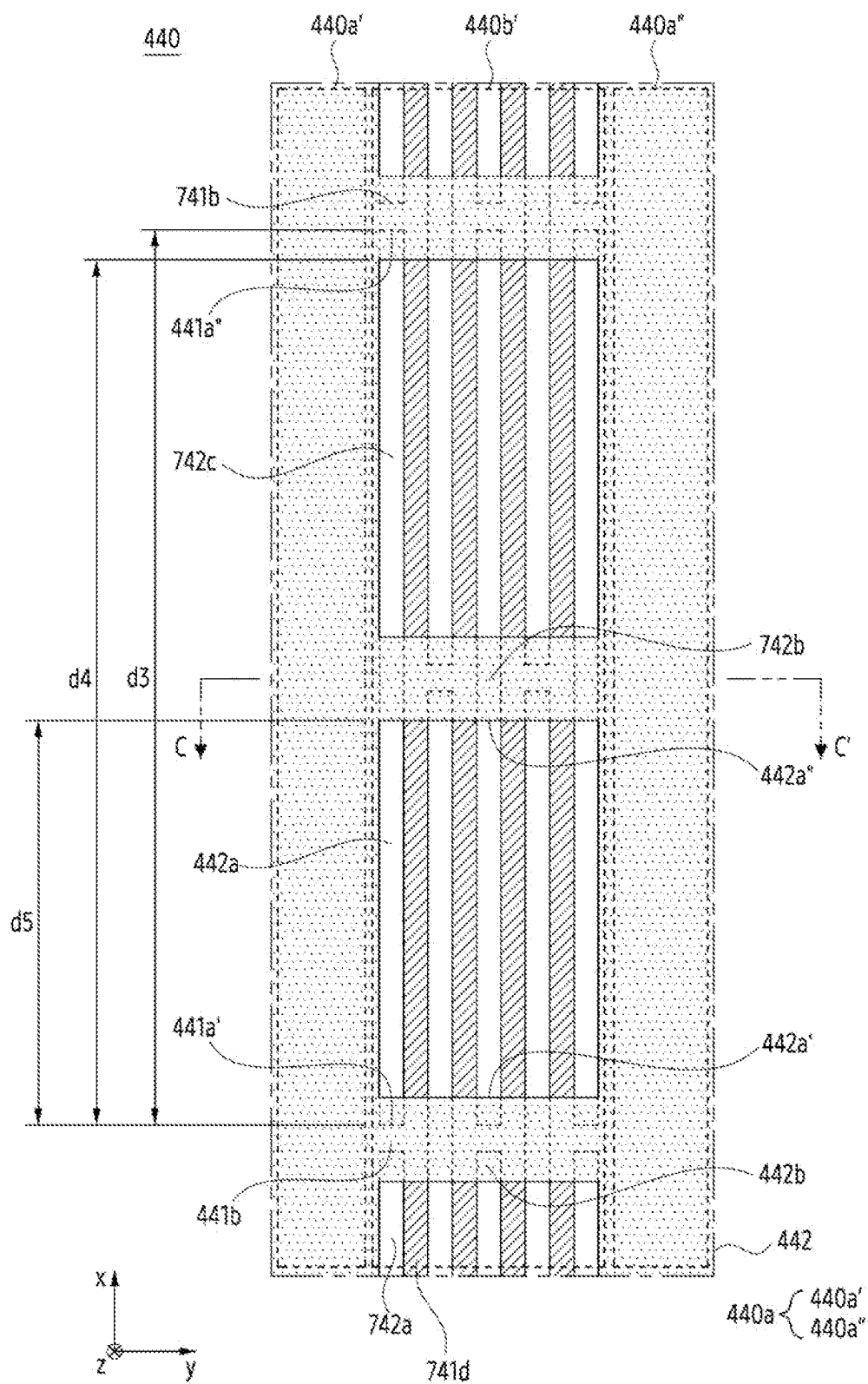


FIG. 9A

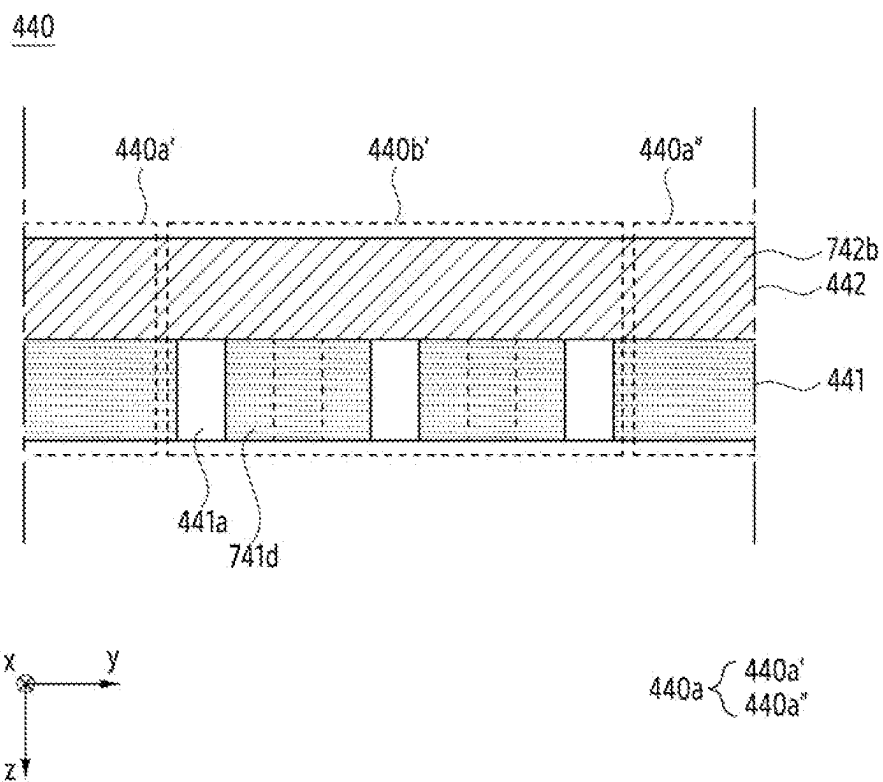


FIG. 9B

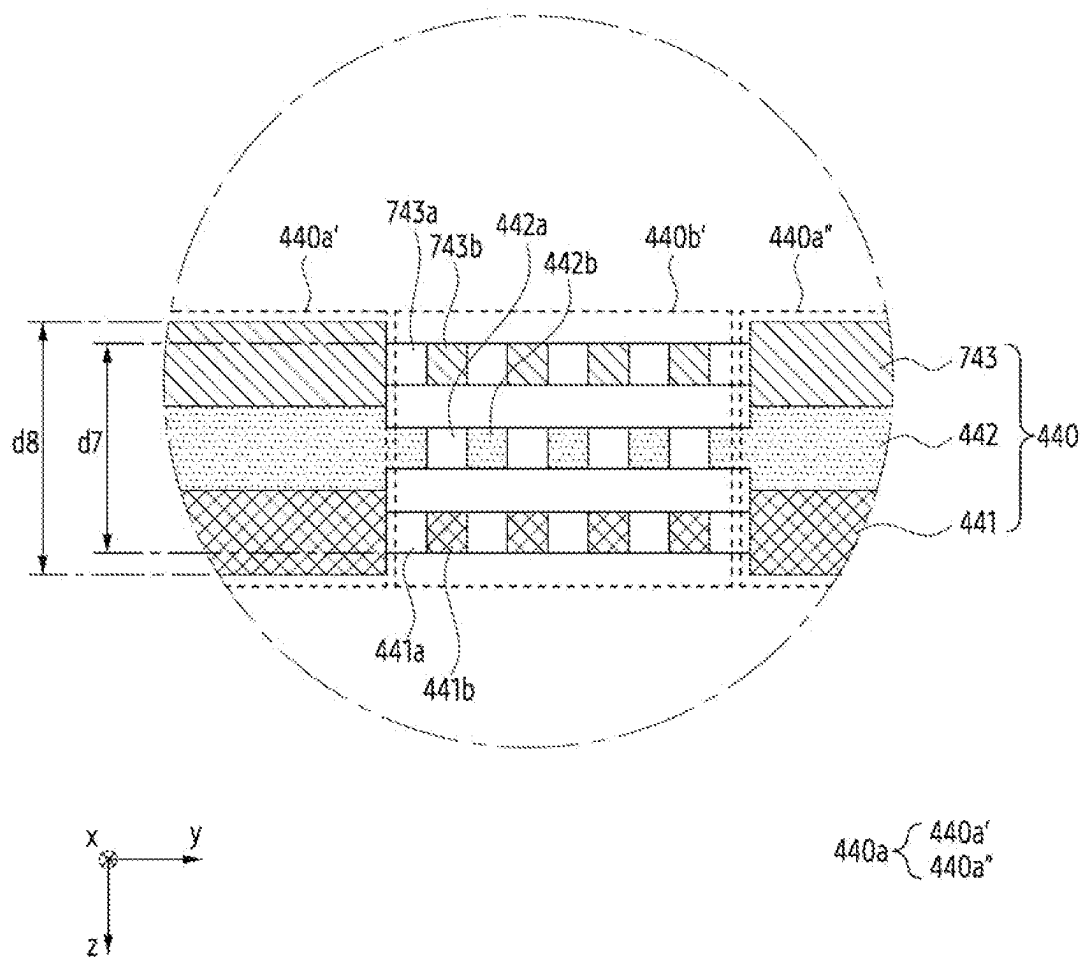


FIG. 10

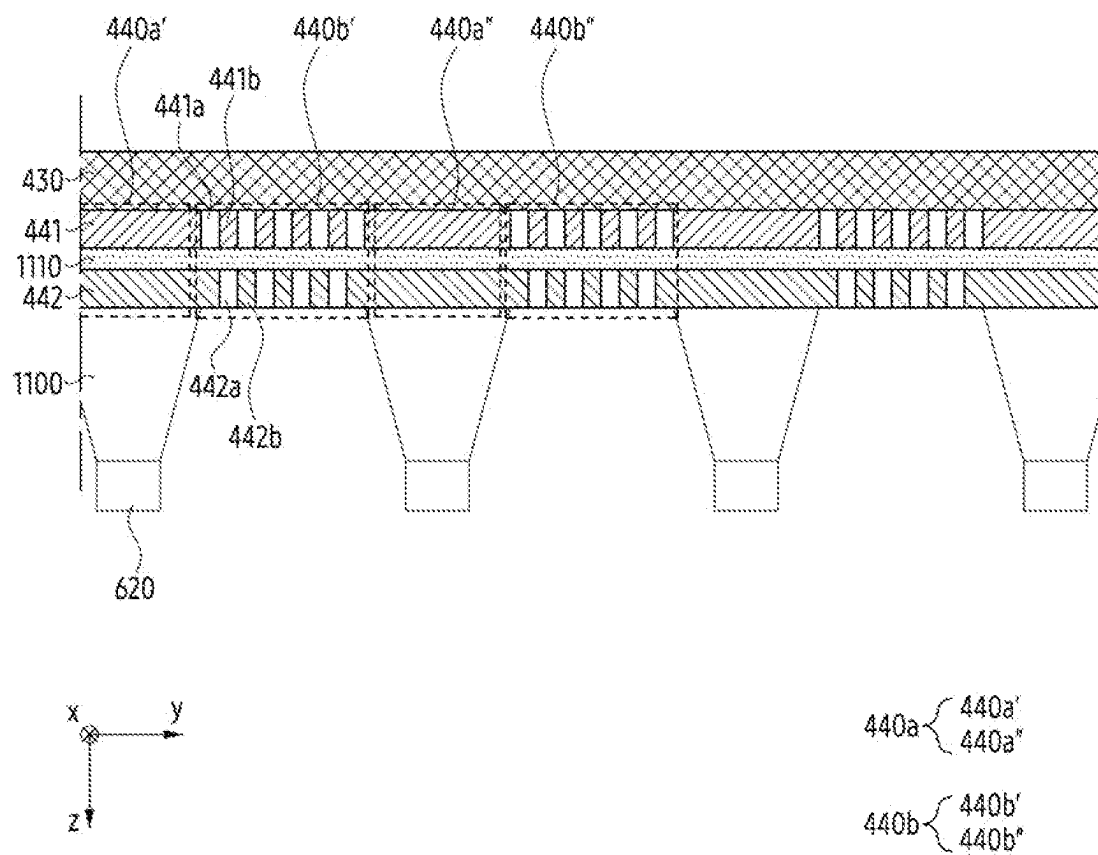


FIG. 11A

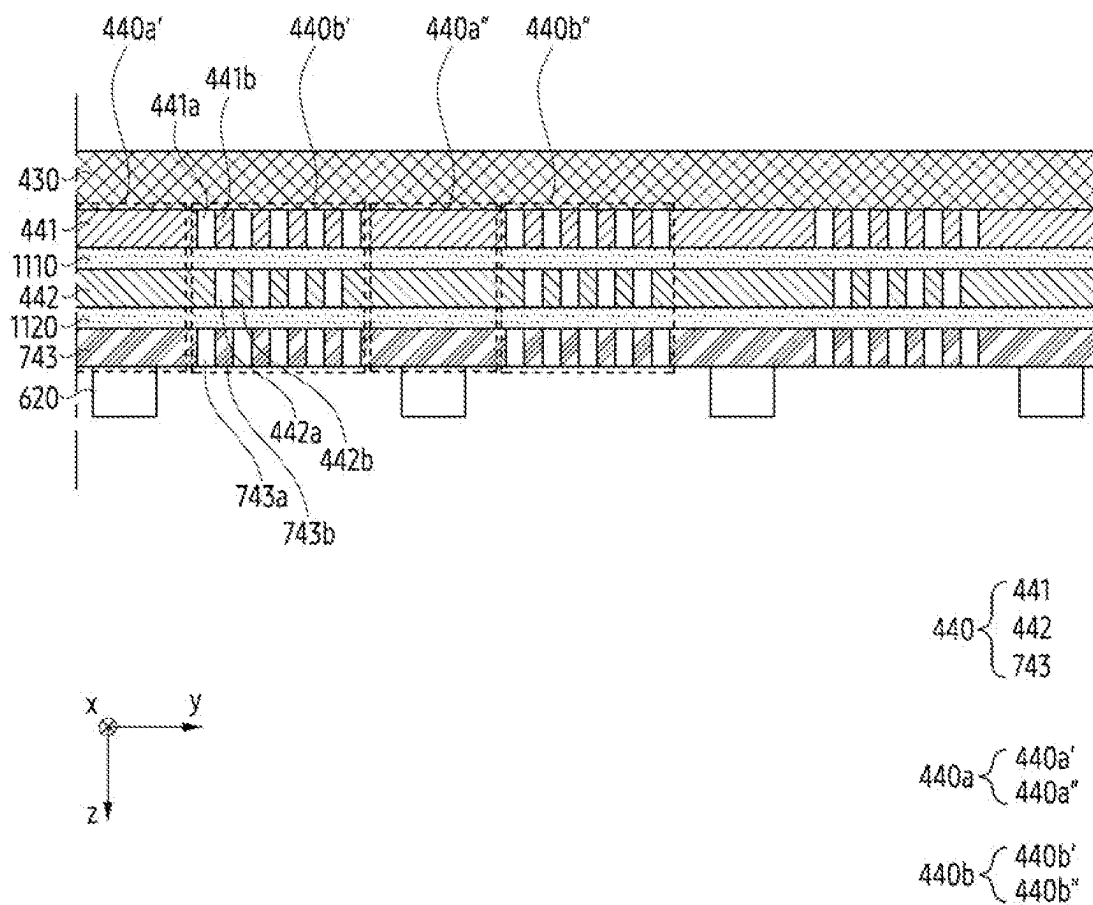


FIG. 11B

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ELECTRONIC DEVICE INCLUDING STRUCTURE FOR SUPPORTING DISPLAY

CROSS-REFERENCE TO RELATED APPLICATION

This application is a by-pass continuation application of International Application No. PCT/KR2023/005181, filed on Apr. 17, 2023, which is based on and claims priority to Korean Patent Application No. 10-2022-0069612, filed on Jun. 8, 2022, and Korean Patent Application No. 10-2022-0100979, filed on Aug. 11, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

The disclosure relates to an electronic device including structure for supporting a display.

2. Description of Related Art

In order for a user to easily carry electronic devices, electronic devices are required to be miniaturized. Notwithstanding such miniaturization of these electronic devices, there is still an increasing need for an electronic device in which a screen size of a display for displaying contents may be changed so that the user can utilize various contents through the electronic device. For example, the electronic device may include a flexible display with a changeable screen size of the display being exposed to the outside of the electronic device.

SUMMARY

In a flexible display of an electronic device, for example, the area of the display may be expanded or reduced according to movement of a housing. In order to reinforce rigidity of the display of which area varies according to the movement of the housing, the electronic device may include a structure capable of supporting the display in the housing. When the thickness of the supporting structure is designed to increase with a view to reinforcing the rigidity of the display, the thickness of the housing may also unnecessarily increase. Thus, the electronic device may need to optimize the supporting structure of the display in the electronic device.

The technical problems to be addressed in this disclosure are not limited to those described above, and other technical problems not mentioned herein will be clearly understood by those having ordinary knowledge in the technical field to which the present disclosure belongs, from the following description.

According to an embodiment, an electronic device includes a first housing, a second housing coupled to the first housing so as to be slidable with respect to the first housing in a first direction and a second direction opposite to the first direction, a display including a planar region disposed on the second housing and a deformable region in contact with the planar region, the deformable region being exposed to an outside as the second housing moves in the first direction or disposed in the first housing as the second housing moves in the second direction, and a support plate supporting the display. According to an embodiment, the support plate includes rigid regions, a flexible region disposed between

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the rigid regions and including a plurality of openings spaced apart from each other and having a length in a third direction perpendicular to the first direction, a first layer including a plurality of first openings extending in the third direction in the flexible region, and a second layer disposed on the first layer and including a plurality of second openings partially overlapping each of the plurality of first openings in the flexible region. According to an embodiment, a distance from one end to the other end of one first opening of the plurality of first openings facing the third direction is different from a distance from the one end of the one first opening to one end of both ends of one second opening of the plurality of second openings overlapping the one first opening, the one second opening being near to the other end of the one first opening.

According to an embodiment, an electronic device includes a first housing, a second housing coupled to the first housing so as to be slidable with respect to the first housing in a first direction and a second direction opposite to the first direction, a display including a planar region disposed on the second housing and a deformable region in contact with the planar region, the deformable region being exposed to an outside as the second housing moves in the first direction or disposed in the first housing as the second housing moves in the second direction, a support plate supporting the display, at least one gear head disposed on both peripheries of the support plate extending in a direction parallel to the first direction, and a guide gear guiding movement of the display and rotating as meshed with the at least one gear. According to an embodiment, the support plate includes rigid regions, a flexible region disposed between the rigid regions, the flexible region including a plurality of openings spaced apart from each other and having a length in a third direction perpendicular to the first direction, a first layer including a plurality of first openings extending in the third direction and a plurality of first bridges separating each of the plurality of first openings in the flexible region, and a second layer disposed on the first layer and including a plurality of second openings partially overlapping each of the plurality of first openings and a plurality of second bridges separating each of the plurality of second openings in the flexible region. According to an embodiment, a distance from one end of one first opening of the plurality of first openings facing the third direction to one first bridge of the plurality of first bridges in contact with another end of the one first opening may be different from a distance from the one end of the one first opening to a second bridge of the plurality of second bridges, the second bridge being near to the another end of the first opening.

An electronic device according to an embodiment, a support plate including a plurality of layers may be disposed onto at least part of a display, thereby reinforcing the display and then providing enough space for other components in the electronic device.

The effects that can be obtained from the disclosure are not limited to those described above, and any other effects not mentioned herein will be clearly understood by those having ordinary knowledge in the technical field to which the disclosure belongs, from the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating an electronic device in a network environment according to various embodiments;

FIG. 2A is a front view illustrating a first state of an electronic device according to an embodiment;

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FIG. 2B is a rear view illustrating a first state of an electronic device according to an embodiment;

FIG. 2C is a front view illustrating a second state of an electronic device according to an embodiment;

FIG. 2D is a rear view illustrating a second state of an electronic device according to an embodiment;

FIG. 3A is an exploded perspective view of an electronic device according to an embodiment;

FIG. 3B is a cross-sectional view illustrating an example of an electronic device according to an embodiment taken along a line A-A' of FIG. 2A;

FIG. 4A is a cross-sectional view of an electronic device in a first state of according to an embodiment;

FIG. 4B is a cross-sectional view of an electronic device in a second state according to an embodiment;

FIGS. 5A and 5B are front views of a display structure of an electronic device according to an embodiment;

FIG. 6A illustrates part of an internal structure of an electronic device with a gear head being disposed according to an embodiment;

FIGS. 6B and 6C are front views of a display structure of an electronic device with a gear head being disposed according to an embodiment;

FIG. 7A is an exploded perspective view of a display support structure of an electronic device according to an embodiment;

FIGS. 7B, 7C, 7D and 7E are front views of a display support structure of an electronic device according to an embodiment;

FIG. 8A is a front view of a display support structure of an electronic device according to an embodiment;

FIG. 8B is a cross-sectional view illustrating an example of a display support structure of an electronic device taken along a line B-B' of FIG. 8A, according to an embodiment;

FIG. 9A is a front view of a display support structure of an electronic device according to an embodiment;

FIG. 9B is a cross-sectional view illustrating an example of a display support structure of an electronic device taken along a line C-C' of FIG. 9A, according to an embodiment;

FIG. 10 is a side view of a display support structure of an electronic device according to an embodiment; and

FIGS. 11A and 11B are side views illustrating a display structure of an electronic device according to an embodiment.

DETAILED DESCRIPTION

FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure.

Referring to FIG. 1, an electronic device 101 in a network environment 100 may communicate with an electronic device 102 via a first network 198 (e.g., a short-range wireless communication network), or at least one of an electronic device 104 or a server 108 via a second network 199 (e.g., a long-range wireless communication network). According to an embodiment, the electronic device 101 may communicate with the electronic device 104 via the server 108. According to an embodiment, the electronic device 101 may include a processor 120, a memory 130, an input module 150, a sound output module 155, a display module 160, an audio module 170, a sensor module 176, an interface 177, a connecting terminal 178, a haptic module 179, a camera module 180, a power management module 188, a battery 189, a communication module 190, a subscriber identification module (SIM) 196, or an antenna module 197. In some embodiments of the disclosure, at least one of the

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components (e.g., the connecting terminal 178) may be omitted from the electronic device 101, or one or more other components may be added in the electronic device 101. In some embodiments of the disclosure, some of the components (e.g., the sensor module 176, the camera module 180, or the antenna module 197) may be implemented as a single component (e.g., the display module 160).

The processor 120 may execute, for example, software (e.g., a program 140) to control at least one other component (e.g., a hardware or software component) of the electronic device 101 coupled with the processor 120, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor 120 may store a command or data received from another component (e.g., the sensor module 176 or the communication module 190) in a volatile memory 132, process the command or the data stored in the volatile memory 132, and store resulting data in a non-volatile memory 134. According to an embodiment, the processor 120 may include a main processor 121 (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor 123 (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor 121. For example, when the electronic device 101 includes the main processor 121 and the auxiliary processor 123, the auxiliary processor 123 may be adapted to consume less power than the main processor 121, or to be specific to a specified function. The auxiliary processor 123 may be implemented as separate from, or as part of the main processor 121.

The auxiliary processor 123 may control at least some of functions or states related to at least one component (e.g., the display module 160, the sensor module 176, or the communication module 190) among the components of the electronic device 101, instead of the main processor 121 while the main processor 121 is in an inactive (e.g., sleep) state, or together with the main processor 121 while the main processor 121 is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor 123 (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module 180 or the communication module 190) functionally related to the auxiliary processor 123. According to an embodiment, the auxiliary processor 123 (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device 101 where the artificial intelligence is performed or via a separate server (e.g., the server 108). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

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The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wired) or wirelessly coupled with the electronic device **101**.

The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

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The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mm Wave band) to address, e.g., a high data transmission rate. The wireless communication module **192** may support vari-

ous technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large-scale antenna. The wireless communication module 192 may support various requirements specified in the electronic device 101, an external electronic device (e.g., the electronic device 104), or a network system (e.g., the second network 199). According to an embodiment, the wireless communication module 192 may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

The antenna module 197 may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device 101. According to an embodiment, the antenna module 197 may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module 197 may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network 198 or the second network 199, may be selected, for example, by the communication module 190 (e.g., the wireless communication module 192) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module 190 and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module 197.

According to various embodiments, the antenna module 197 may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

According to an embodiment, commands or data may be transmitted or received between the electronic device 101 and the external electronic device 104 via the server 108 coupled with the second network 199. Each of the electronic devices 102 or 104 may be a device of a same type as, or a different type, from the electronic device 101. According to an embodiment, all or some of operations to be executed at the electronic device 101 may be executed at one or more of the external electronic devices 102, 104, or 108. For example, if the electronic device 101 should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device 101, instead of, or in addition to, executing the function or

the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device 101. The electronic device 101 may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device 101 may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment of the disclosure, the external electronic device 104 may include an Internet-of-Things (IoT) device. The server 108 may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device 104 or the server 108 may be included in the second network 199. The electronic device 101 may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

FIG. 2A is a front view of an electronic device in a first state according to an embodiment, FIG. 2B is a rear view of the first state of the electronic device according to an embodiment, FIG. 2C is a front view of a second state of the electronic device according to an embodiment, and FIG. 2D is a rear view of the second state of the electronic device according to an embodiment.

Referring to FIGS. 2A, 2B, 2C, and 2D, an electronic device 200 (e.g., the electronic device 101 of FIG. 1) according to an embodiment may include a first housing 210, a second housing 220, a display 230 (e.g., the display module 160 of FIG. 1), and a camera 240 (e.g., the camera module 180 of FIG. 1). According to an embodiment, the second housing 220 may be slidable with respect to the first housing 210. For example, the second housing 220 may move within a range of a specified distance along a first direction (e.g., +y direction) with respect to the first housing 210. When the second housing 220 moves in the first direction, a distance between a side surface 220a of the second housing 220 facing the first direction and the first housing 210 may increase. For another example, the second housing 220 may move within a range of a specified distance along a second direction (e.g., -y direction) opposite to the first direction with respect to the first housing 210. When the second housing 220 moves in the second direction, the distance between the side surface 220a of the second housing 220 facing the first direction and the first housing 210 may decrease. According to an embodiment, the second housing 220 may linearly reciprocate with respect to the first housing 210 while sliding relative to the first housing 210. For example, at least a portion of the second housing 220 may be slidable into the first housing 210 or slidable out from the first housing 210.

According to an embodiment, the electronic device 200 may be referred to as a “slidable electronic device”, as the second housing 220 is designed to be slidable with respect to the first housing 210. According to an embodiment, the electronic device 200 may be referred to as a “rollable electronic device”, as at least a portion of the display 230 is designed to be rolled up into the second housing 220 (or the first housing 210) based on a sliding movement of the second housing 220.

According to an embodiment, the first state of the electronic device 200 may be defined as a state in which the

second housing 220 is moved in the second direction (e.g., a contracted state or a slide-in state). For example, in the first state of the electronic device 200, the second housing 220 may be movable in the first direction, but may not be movable in the second direction. In the first state of the electronic device 200, the distance between the side surface 220a of the second housing 220 and the first housing 210 may increase as the second housing 220 moves, but it may not decrease any more. For another example, in the first state of the electronic device 200, a portion of the second housing 220 may be slidable-out from the first housing 210, but it may not be slidable-in anymore. According to an embodiment, the first state of the electronic device 200 may be defined as a state in which a second region 230b of the display 230 is not visually exposed from the outside of the electronic device 200. For example, in the first state of the electronic device 200, the second region 230b of the display 230 may be located in an internal space of the electronic device 200 formed by the first housing 210 and/or the second housing 220, so that it is not visible from the outside of the electronic device 200.

According to an embodiment, the second state of the electronic device 200 may be defined as a state in which the second housing 220 is moved in the first direction (e.g., an extended state or a slide-out state). For example, in the second state of the electronic device 200, the second housing 220 may be movable in the second direction, but may not be movable in the first direction anymore. In the second state of the electronic device 200, the distance between the side surface 220a of the second housing 220 and the first housing 210 may decrease as the second housing 220 moves, but it may not increase further. For another example, in the second state of the electronic device 200, a portion of the second housing 220 may be slidable into the first housing 210, but it may not be slidable out from the first housing 210 further. According to an embodiment, the second state of the electronic device 200 may be defined as a state in which the second region 230b of the display 230 is visually exposed from the outside of the electronic device 200. For example, in the second state of the electronic device 200, the second region 230b of the display 230 may be drawn out from the internal space of the electronic device 200 so that it is visible from the outside of the electronic device 200.

According to an embodiment, when the second housing 220 moves from the first housing 210 in the first direction, at least a portion of the second housing 220 and/or the second region 230b of the display 230 may be drawn out from the first housing 210 by a pull-out length d1 corresponding to the moving distance of the second housing 220. According to an embodiment, the second housing 220 may reciprocate within a specified distance d2. According to an embodiment, the pull-out length d1 may have substantially a dimension ranging from zero to the specified distance d2.

According to an embodiment, the state of the electronic device 200 may be convertible between the second state and/or the first state, with either a manual operation by a user or an automatic operation by a driving module arranged inside the first housing 210 or the second housing 220. According to an embodiment, the driving module may trigger a certain operation based on a user input. According to an embodiment, the user input for triggering the operation of the driving module may include a touch input through the display 230, a force touch input, and/or a gesture input. According to another embodiment, the user input for triggering the operation of the driving module may include a voice input or an input of a physical button exposed to the outside of the first housing 210 or the second housing 220.

According to an embodiment, the driving module may be configured to be driven in a semi-automatic manner in which an operation is triggered upon detection of such a manual operation using a user's physical force.

According to an embodiment, the first state of the electronic device 200 may be referred to as a "first shape", and the second state of the electronic device 200 may be referred to as a "second shape". For example, the first shape may include a normal state, a reduced state, or a closed state, and the second shape may include an open state. According to an embodiment, the electronic device 200 may form a third state (e.g., an intermediate state) located between the first state and the second state. For example, the third state may be referred to as a "third shape", and the third shape may include a free stop state.

According to an embodiment, the display 230 may be visible or viewable from the outside in a front direction (e.g., -z direction) of the electronic device 200 so as to display visual information to the user. For example, the display 230 may include a flexible display. According to an embodiment, the display 230 may be disposed in the second housing 220 and be pulled-out from the internal space of the electronic device 200 according to the movement of the second housing 220 or may be pulled-in into the internal space of 200 of the electronic device. The internal space of the electronic device 200 may refer to an internal space defined by the first housing 210 and the second housing 220 formed by coupling of the first housing 210 and the second housing 220. For example, in the first state of the electronic device 200, at least a portion of the display 230 may be rolled up to be slid into the internal space of the electronic device 200. When the second housing 220 moves in the first direction while at least a portion of the display 230 is slidably inserted into the internal space of the electronic device 200, at least a portion of the display 230 may be drawn out from the internal space of the electronic device 200. For another example, when the second housing 220 moves in the second direction, at least a portion of the display 230 may be rolled up into the inside of the electronic device 200 to be inserted into the internal space of the electronic device 200. As at least the portion of the display 230 is slid out from the internal space of the electronic device 200 or slid into the internal space of the electronic device 200, the display area of the display 230 visible from the outside of the electronic device 200 may be expanded or reduced. According to an embodiment, the display 230 may include a first region 230a and a second region 230b.

According to an embodiment, the first region 230a of the display 230 may fixedly refer to an area of the display 230 that is visible from the outside of the electronic device 200 regardless of whether the electronic device 200 is in the second state or the first state. For example, the first region 230a may indicate a partial area of the display 230 that is not rolled up into the internal space of the electronic device 200. According to an embodiment, as the second housing 220 moves, the first region 230a may move together with the second housing 220. For example, when the second housing 220 moves in the first direction or in the second direction, the first region 230a may be moved along with the second housing 220 on a front surface of the electronic device 200 either in the first direction or in the second direction.

According to an embodiment, the second region 230b of the display 230 may be connected to the first region 230a, and may be slid into the internal space or slid out from the internal space of the electronic device 200 as the second housing 220 moves. For example, the second region 230b of the display 230 may be in a rolled-up state, being retracted

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into the internal space of the electronic device **200**, in the first state of the electronic device **200**. The second region **230b** of the display **230** may not be visible from the outside as it is fully slid into the internal space of the electronic device **200** in the first state of the electronic device **200**. As another example, the second region **230b** of the display **230** may be in a state drawn out from the internal space of the electronic device **200** in the second state of the electronic device **200**. In the second state, the second region **230b** of the display **230** may be visible from the outside of the electronic device **200**.

According to an embodiment, in the first state of the electronic device **200**, the area of the display **230** that is visible from the outside of the electronic device **200** may include only the first region **230a** of the display **230**. In the second state of the electronic device **200**, the area of the display **230** that is visible from the outside of the electronic device **200** may include the first region **230a** and at least a portion of the second region **230b** of the display **230**.

According to an embodiment, the first housing **210** of the electronic device **200** may include a book cover **211** surrounding the internal space of the first housing **210** and a rear plate **212** surrounding a back surface of the book cover **211**. The second housing **220** of the electronic device **200** may include a front cover **221** surrounding the internal space of the electronic device **200**.

According to an embodiment, the front cover **221** may include a first cover area **221a** of the front cover **221** that is not pulled-in into the interior of the first housing **210**, and a second cover area **221b** that is pulled-in or pulled-out into/from the interior of the first housing **210**. The first cover area **221a** of the front cover **221** may be always visible regardless of whether the electronic device **200** is either in the second state or in the first state. According to an embodiment, at least a portion of the first cover area **221a** of the front cover **221** may form a side surface **220a** of the second housing **220**. According to an embodiment, the second cover area **221b** of the second housing **220** may not be visible in the first state, while it may be visible in the second state.

The camera **240** may acquire an image of a subject based on receiving light from the outside of the electronic device **200**. According to an embodiment, the camera **240** may include one or more lenses, an image sensor, and/or an image signal processor. According to an embodiment, the camera **240** may be disposed on the second housing **220** so that it faces the rear surface of the electronic device **200** opposite to the front surface of the electronic device **200** on which the first region **230a** of the display **230** is disposed. For example, the camera **240** may be disposed on the front cover **221** of the second housing **220** and may be visible from the outside of the electronic device **200** through an opening **211a** formed in the book cover **211**, when the electronic device **200** is in the first state. As another example, the camera **240** may be disposed on the front cover **221** of the second housing **220**, and when the electronic device **200** is in the first state, it may be obscured by the book cover **211** and/or the rear plate **212**, so that it may not be visible from the outside of the electronic device **200**.

According to an embodiment, the camera **240** may include a plurality of cameras. For example, the camera **240** may include a wide-angle camera, an ultra-wide-angle camera, a telephoto camera, a close-up camera, and/or a depth camera. However, the camera **240** is not necessarily limited to including a plurality of cameras, and may include a single camera.

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According to an embodiment, the camera **240** may further include a camera facing the front surface of the electronic device **200** on which the first region **230a** of the display **230** is disposed. When the camera **240** faces the front surface of the electronic device **200**, the camera **240** may be an under-display camera (UDC) disposed below the display **230** (e.g., in +z direction from the display **230**), but the disclosure is not limited thereto.

According to an embodiment, the electronic device **200** may include a sensor module and/or a camera module disposed underneath the display **230**. The sensor module may detect an external environment based on information (e.g., light) received through the display **230**. According to an embodiment, the sensor module may include at least one of a receiver, a proximity sensor, an ultrasonic sensor, a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an IR (infrared) sensor, a biometric sensor, a temperature sensor, a humidity sensor, a motor encoder or an indicator. According to an embodiment, at least some sensor modules of the electronic device **200** may be visually exposed to the outside through a partial area of the display **230**. According to an embodiment, the electronic device **200** may detect a pull-out length using a sensor module. According to an embodiment, the electronic device **200** may generate information on the pull-out length detected by the sensor. For example, the electronic device **200** may detect and/or identify a degree of pull-out of the second housing **220** using the pull-out information. According to an embodiment, the pull-out information may include information on the pull-out length of the second housing **220**.

According to an embodiment, such a shape of coupling of the first housing **210** and the second housing **220** is not limited to the shape of coupling illustrated in FIGS. **2A**, **2B**, **2C** and **2D**, and it may be implemented by a combination and/or coupling of other shapes or parts.

FIG. **3A** is an exploded perspective view of an electronic device according to an embodiment, and FIG. **3B** is a cross-sectional view of an electronic device according to an embodiment, taken along a line A-A' of FIG. **2A**.

Referring to FIGS. **3A** and **3B**, according to an embodiment, the electronic device **200** may include a first housing **210**, a second housing **220**, a display **230**, a camera **240**, a battery **250** (e.g., the battery **189** of FIG. **1**) and a driving module **260**. According to an embodiment, the first housing **210** and the second housing **220** may be coupled to each other to form an internal space **201** of the electronic device **200**. For example, in the first state of the electronic device **200**, the second region **230b** of the display **230** may be accommodated in the internal space **201**.

According to an embodiment, the first housing **210** may include a book cover **211**, a rear plate **212**, and a frame cover **213**. According to an embodiment, the book cover **211**, the rear plate **212** and the frame cover **213** arranged in the first housing **210** may be coupled to each other, so that they may not move together with the second housing **220** when the second housing **220** moves with respect to the first housing **210**.

According to an embodiment, the book cover **211** may form at least a portion of an outer surface of the electronic device **200**. For example, the book cover **211** may form at least a part of the side surface of the electronic device **200** and at least a part of the rear surface of the electronic device **200**. According to an embodiment, the book cover **211** may provide a surface on which the rear plate **212** is seated. The rear plate **212** may be seated on one surface **211b** of the book cover **211**.

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According to an embodiment, the frame cover **213** may be configured to support various components within the electronic device **200**. For example, the frame cover **213** may be configured to accommodate at least a portion of the battery **250** and the driving module **260**. The battery **250** and the driving module **260** may be accommodated in at least one of a recess or a hole provided in the frame cover **213**. According to an embodiment, the frame cover **213** may be surrounded by the book cover **211**. For example, in the first state of the electronic device **200**, one surface **213a** of the frame cover **213** on which the battery **250** is disposed may be arranged to face the book cover **211** and/or the second region **230b** of the display **230**. As another example, in the first state of the electronic device **200**, the other surface **213b** of the frame cover **213** facing the one surface **213a** of the frame cover **213** may be arranged to face the first region **230a** of the display **230** or the front cover **221**. For example, the frame cover **213** may include an aluminum material, but is not limited thereto.

According to an embodiment, the second housing **220** may include a front cover **221**, a rear cover **222** and a slide cover **223**. According to an embodiment, the front cover **221**, the rear cover **222** and the slide cover **223** may be coupled to each other, and when the second housing **220** moves relative to the first housing **210**, it may also move together with the second housing **220**. The front cover **221** may be configured to support various components within the electronic device **200**. For example, the camera **240** may be disposed on one surface **221c** of the front cover **221** facing the internal space **201**. The other surface **221d** of the front cover **221** facing the one surface **221c** of the front cover **221** may be arranged to face the first region **230a** of the display **230**, when the electronic device **200** is in the first state. According to an embodiment, the rear cover **222** may be coupled to the front cover **221** to protect various components in the electronic device **200** disposed on the front cover **221**. For example, the rear cover **222** may cover a portion of one surface **221c** of the front cover **221**. According to an embodiment, the slide cover **223** may be disposed on the rear cover **222** to form an outer surface of the electronic device **200** together with the rear plate **212** and the book cover **211**. The slide cover **223** may be coupled to one surface of the rear cover **222** to protect the rear cover **222** and/or the front cover **221**.

According to an embodiment, when the electronic device **200** is in the first state, at least a part of the display **230** may be rolled up into the internal space **201**, thereby causing it to be bent. According to an embodiment, the display **230** may cover at least a portion of the frame cover **213** and at least a portion of the front cover **221**. For example, when the electronic device **200** is in the first state, the display **230** may be configured to cover the other surface **221d** of the front cover **221** and pass between the front cover **221** and the book cover **211** to be extended toward the internal space **201**. When the electronic device **200** is in the first state, the display **230** may pass between the front cover **221** and the book cover **211** and surround the frame cover **213** disposed in the internal space **201**. The display **230** may cover one surface **213a** of the frame cover **213** in the internal space **201**. According to an embodiment, when the second housing **220** moves in the first direction, the second region **230b** of the display **230** may be pulled-out from the internal space **201**. For example, as the second housing **220** moves in the first direction, the display **230** may pass between the front cover **221** and the book cover **211** and may be pulled-out from the internal space **201**.

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According to an embodiment, the first region **230a** of the display **230** may be in contact with the other surface **221d** of the front cover **221**. For example, the first region **230a** may extend parallel to the other surface **221d** of the front cover **221** while coming into contact with the other surface **221d** of the front cover **221**. Due to extending parallel to the other surface **221d** of the front cover **221**, the first region **230a** may have a substantially planar shape. According to an embodiment, the first region **230a** of the display **230** may not be deformed according to movement of the second housing **220**. For example, the first region **230a** may move along the movement of the second housing **220** while keeping such a planar shape.

According to an embodiment, the second region **230b** of the display **230** may be deformable according to the movement of the second housing **220**. For example, when the electronic device **200** is in the first state, the second region **230b** may be bent with a certain curvature in the internal space **201** of the electronic device **200**. When the second housing **220** moves in the first direction (e.g., +y direction), at least a portion of the second region **230b** may be pulled out from the internal space **201** of the electronic device **200** to be parallel to the other surface **221d** of the front cover **221**. When at least a portion of the second region **230b** is parallel to the other surface **221d** of the front cover **221**, at least a portion of the second region **230b** may have no curvature and have a planar shape. As another example, when the second housing **220** moves in the second direction (e.g., -y direction), the at least a portion of the second region **230b** may be slid into the internal space **201** of the electronic device **200**. The at least a portion of the second region **230b** may be rolled up to be bent with a certain curvature while being pulled-in into the internal space **201** of the electronic device **200**.

According to an embodiment, the electronic device **200** may include a support member **231** supporting the display **230** and a guide rail **232**. The support member **231** may include a plurality of bars coupled to each other, and may be manufactured in a shape corresponding to the shape of the second region **230b** of the display **230**. For example, the support member **231** may include a plurality of bars extending in a third direction (e.g., +x direction) perpendicular to a first direction (e.g., a +y direction). The support member **231** may be disposed on the second region **230b** of the display **230**. According to an embodiment, the support member **231** may move along with the display **230** as the display **230** moves. According to an embodiment, in the first state that the second region **230b** of the display **230** is rolled up within the internal space **201**, the support member **231** may be rolled up inside the internal space **201** together with the second region **230b** of the display **230**.

According to an embodiment, the guide rail **232** may be configured to guide the movement of the support member **231**. For example, as the display **230** moves, the support member **231** may move along the guide rail **232** coupled to the frame cover **213**. According to an embodiment, the guide rail **232** may include a plurality of guide rails **232** that are arranged spaced apart from each other at both peripheries of the frame cover **213** being spaced apart from each other in the third direction (e.g., +x direction) perpendicular to the first direction.

According to an embodiment, the driving module **260** may provide a driving force to the second housing **220** so that the second housing **220** can move relative to the first housing **210**. According to an embodiment, the driving module **260** may include a motor **261**, a pinion gear **262**, a rack gear **263** and a bracket **264**. The motor **261** may receive

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power from the battery 250 to provide the driving force to the second housing 220. According to an embodiment, the motor 261 may be disposed in the first housing 210 and, when the second housing 220 moves with respect to the first housing 210, it may not move together with the second housing 220. For example, the motor 261 may be disposed in a recess formed in the frame cover 213. According to an embodiment, the pinion gear 262 may be coupled to the motor 261 and rotate by a driving force provided from the motor 261. For example, the pinion gear 262 may be rotatably coupled to a shaft 261a that is a drive shaft of the motor 261, and may be rotatable with respect to the motor 261. According to an embodiment, the rack gear 263 may be meshed with the pinion gear 262 and move according to the rotation of the pinion gear 262. For example, the rack gear 263 may linearly reciprocate in either the first direction or the second direction according to the rotation of the pinion gear 262. According to an embodiment, the rack gear 263 may be disposed in the second housing 220. For example, the rack gear 263 may be coupled to the front cover 221 included in the second housing 220. According to an embodiment, the rack gear 263 may be movable within an operating space 213p formed in the frame cover 213. According to an embodiment, the bracket 264 may maintain a position of the pinion gear 262 so that the driving force of the motor 261 can be seamlessly transmitted to the rack gear 263.

According to an embodiment, when the pinion gear 262 rotates along the first rotational direction (e.g., clockwise in FIG. 4B), the rack gear 263 may move in the first direction (e.g., +y direction). When the rack gear 263 moves in the first direction, the second housing 220 coupled to the rack gear 263 may move in the first direction. As the second housing 220 moves in the first direction, the area of the display 230 that is visible from the outside of the electronic device 200 may be expanded. When the pinion gear 262 rotates in the second rotational direction (e.g., counterclockwise in FIG. 4B), the rack gear 263 may move in the second direction (e.g., -y direction). When the rack gear 263 moves in the second direction, the second housing 220 coupled to the rack gear 263 may also move in the second direction. As the second housing 220 moves in the second direction, the area of the display 230 that is visible from the outside of the electronic device 200 may be reduced.

In the above description, it has been described that the motor 261 and the pinion gear 262 are disposed in the first housing 210, and the rack gear 263 is disposed in the second housing 220, but the present disclosure is not limited thereto. According to other embodiments, the motor 261 and the pinion gear 262 may be disposed in the second housing 220, and the rack gear 263 may be disposed in the first housing 210.

FIG. 4A is a cross-sectional view of an electronic device in a first state according to an embodiment, and FIG. 4B is a cross-sectional view of an electronic device in a second state according to an embodiment.

Referring to FIGS. 4A and 4B, the electronic device 400 (e.g., the electronic device 101 of FIG. 1 or the electronic device 200 of FIG. 2A) may include a first housing 410 (e.g., the first housing 210 of FIG. 2A), a second housing 420 (e.g., the second housing 220 of FIG. 2A), a display 430 (e.g., the display module 160 of FIG. 1 or the display 230 of FIG. 2A), and a support plate 440.

According to an embodiment, the first housing 410 may constitute at least part of an exterior of the electronic device 400. The first housing 410 may include components consti-

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tuting at least part of the exterior of the electronic device 400 (e.g., the book cover 211, the rear plate 212, and the frame cover 213 of FIG. 3B).

According to an embodiment, the second housing 420 may make up the remaining part of the exterior of the electronic device 400 in combination with the first housing 410. The second housing 420 may include components constituting at least part of the exterior of the electronic device 400 (e.g., the front cover 221, the rear cover 222, and the slide cover 223 of FIG. 3B).

According to an embodiment, the second housing 420 may be coupled to the first housing 410 so as to be slidable with respect to the first housing 410 along a first direction (e.g., +y direction) or along a second direction (e.g., -y direction) opposite to the first direction. For example, the second housing 420 may be coupled to at least a portion of the display 430. The second housing 420 may expand the display 430, moving along the first direction (+y direction). The second housing 420 may move along the second direction opposite to the first direction, thereby reducing the display 430.

According to an embodiment, the display 430 may make up at least part of the exterior of the electronic device 400. At least a portion of the display 430 may be exposed to the outside of the electronic device 400. For example, the display 430 may be supported by at least a portion of the first housing 410 and/or at least a portion of the second housing 420. One surface of the display 430 may be exposed to the outside of the electronic device 400.

According to an embodiment, the display 430 may include a planar region 430a (e.g., the first area 230a of FIG. 3B) and a deformable region 430b (the second area 230b of FIG. 3B). The planar region 430a may be an area disposed on the second housing 420 and exposed to the outside of the electronic device 400. The deformable region 430b may be, in the first state, pulled-in to be accommodated inside the electronic device 400, and may be, in the second state, pulled-out to be exposed to the outside of the electronic device 400. For example, the planar region 430a may be an area that may be always exposed to the outside of the electronic device 400, and the deformable region 430b may be an area that can be either expanded or reduced according to the state of the electronic device 400.

According to an embodiment, the deformable region 430b may be in contact with the planar region 430a. The deformable region 430b may extend from the planar region 430a. The deformable region 430b may be exposed to the outside as the second housing 420 moves in the first direction (e.g., +y direction) or may be positioned within the first housing 410 as the second housing 420 moves in the second direction (e.g., -y direction). For example, in the first state, the planar region 430a may be exposed to the outside, being coupled with a part of the second housing 420. The deformable region 430b may be in the first state positioned in an internal space of the electronic device 400, being rolled into the electronic device 400. The driving module of the electronic device 400 (e.g., the driving module 260, the motor 261, the pinion gear 262, the rack gear 263, and/or the bracket 264 of FIG. 3A) may move the second housing 420 in the first direction (e.g., +y direction) with respect to the first housing 410. For example, the driving module may transmit power to the second housing 420. Using the transmitted power, the planar region 430a supported by the second housing 420 may move in the first direction. Since the deformable region 430b is in contact with the planar region 430a and extends from the planar region 430a, the deformable region 430b may be exposed to the outside in the second state.

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According to an embodiment, the support plate **440** may support the display **430**. The support plate **440** may be disposed on one surface of the display **430** facing the internal space of the first housing **410** and the second housing **420** (e.g., in +z direction). The support plate **440** may be supported by at least a portion of the second housing **420**. At least a portion of the support plate **440** may be disposed between the second housing **420** and the display **430**.

According to an embodiment, the shape of the support plate **440** may correspond to the shape of the display **430** in order to support the display **430**. The support plate **440** may be configured to change in shape according to the deformation of the display **430**. For example, at least a portion of the support plate **440** may be disposed on the planar region **430a** of the display **430**. At least a portion of the support plate **440** may have a planar shape corresponding to the shape of the planar region **430a**. For example, at least a portion of the support plate **440** may be disposed on the deformable region **430b** of the display **430**. The deformable region **430b** may be in the first state rolled up into the internal space of the electronic device **400** and be in the second state deformed into the planar shape to be exposed to the outside of the electronic device **400**. A portion of the support plate **440** disposed on the deformable region **430b** may be configured to correspond to the deformed shape. For example, the deformable region **430b** may come into contact with the planar region **430a** and extend from the planar region **430a**. The support plate **440** may have a shape corresponding to the planar region **430a** and the deformable region **430b**, so as to be positioned on the planar region **430a** and the deformable region **430b**.

According to an embodiment, the support plate **440** may include at least one rigid region **440a** and at least one flexible region **440b** extending from the rigid region **440a**. The rigid region **440a** may support the display **430**. The flexible region **440b** may be supported by the rigid region **440a**. The flexible region **440b** may be configured to change the shape of the support plate **440** to a shape corresponding to the shape according to the deformation of the display **430**. The support plate **440** may include the rigid region **440a** and the flexible region **440b** so that it may be configured to support the display **430** and be changed to a shape corresponding to the display **430**.

According to an embodiment, the support plate **440** may include rigid regions **440a'** and **440a''** and flexible regions **440b'** and **440b''** disposed between the rigid regions **440a'** and **440a''**. For example, the first rigid region **440a'** may be connected to one end of the first flexible region **440b'** in the first direction. The second rigid region **440a''** may be connected to the other end of the first flexible region **440b'** in the second direction (e.g., -y direction). For example, the first flexible region **440b'** may be disposed between the first rigid region **440a'** and the second rigid region **440a''**. The first rigid region **440a'** and the second rigid region **440a''** may be spaced apart from each other with the first flexible region **440b'** being interposed therebetween. One side of the second flexible region **440b''** may be in contact with the second rigid region **440a''**, and the other side of the second flexible region **440b''** may be in contact with another rigid region distinct from the first rigid region **440a'** and the second rigid region **440a''**. Each of the flexible regions **440b'** and **440b''** may be arranged in a space between the rigid regions **440a'** and **440a''**.

Heretofore, the support plate **440** has been described as including the first rigid region **440a'**, the second rigid region **440a''**, and the first flexible region **440b'**, but the disclosure

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is not limited thereto. The support plate **440** may include a plurality of rigid regions and a plurality of soft regions arranged between the plurality of rigid regions. For example, the support plate **440** may include the rigid regions **440a'** and **440a''**, and the flexible regions **440b'** and **440b''**. The first flexible region **440b'** may be disposed between the first rigid region **440a'** and the second rigid region **440a''**. The second rigid region **440a''** may be disposed between the first flexible region **440b'** and the second flexible region **440b''**. The first rigid region **440a'**, the first flexible region **440b'**, the second rigid region **440a''**, and the second flexible region **440b''** may be sequentially arranged along the first direction (e.g., +y direction).

According to an embodiment, the support plate **440** may include a first layer **441** and a second layer **442**. The first layer **441** may be disposed on one surface facing the internal space of the electronic device **400** to support the display **430**. The first layer **441** may be disposed between the display **430** and the second layer **442**. A portion of the first layer **441** may overlap the planar region **430a** of the display **430**, when looking at the planar region **430a** of the display **430**. The remaining part of the first layer **441** may overlap the deformable region **430b** of the display **430**, when looking at the deformable region **430b** of the display **430**.

The second layer **442** may be disposed on the first layer **441**. A portion of the second layer **442** may be disposed between the second housing **420** and the first layer **441**. A portion of the second layer **442** may overlap the planar region **430a** of the display **430** when looking at the planar region **430a** of the display **430**. The remaining part of the second layer **442** may overlap the deformable region **430b** of the display **430**, when looking at the deformable region **430b** of the display **430**.

According to an embodiment, the support plate **440** may include at least one of carbon fiber reinforced polymer (CFRP), glass fiber reinforced plastics (GFRP), or stainless used steel (SUS). For example, the first layer **441** and/or the second layer **442** included in the support plate **440** may include at least one of CFRP, GFRP, or SUS.

According to an embodiment, the first layer **441** and the second layer **442** may be configured to be deformed into a shape corresponding to a changed shape of the display **430**. For example, each of the layers **441** and **442** may include a plurality of rigid regions and flexible regions disposed between the plurality of rigid regions. Each of the flexible regions included in the first layer **441** may be configured to overlap each of the flexible regions included in the second layer **442**, when looking at the support plate **440**. The flexible regions included in the first layer **441** and the second layer **442** may have at least one opening or groove. The openings or grooves included in the flexible regions may provide flexibility for deformation of the first layer **441** and the second layer **442**. For example, one of the plurality of flexible regions included in the first layer **441** and one of the plurality of flexible regions included in the second layer **442** may overlap each other to form the first flexible region **440b'** of the support plate **440**. As the layers **441** and **442** include overlapped flexible regions, a portion of the layers **441** and **442** disposed on the deformable region **430b** of the display **430** may be configured to cause the display **430** to be deformed to a shape corresponding to the shape of the deformable region **430b** as it moves from the first state to the second state or from the second state to the first state.

According to an embodiment, each of the plurality of rigid regions included in the layers **441** and **442** may be configured to overlap each other. For example, one of the plurality of rigid regions included in the first layer **441** and one of the

plurality of rigid regions included in the second layer **442** can be overlapped to make up the first rigid region **440a'** of the support plate **440**. The overlapped rigid regions included in the layers **441** and **442** may support the display **430**.

According to the above-described embodiment, the electronic device **400** includes the first layer **441** and the second layer **442** in the support plate **440** supporting the display **430** to support the display **430**. The first layer **441** and the second layer **442** includes a plurality of flexible regions so that it can be deformed to a shape corresponding to a shape according to the deformation of the display **430**.

FIGS. **5A** and **5B** each show a front view of a display structure of an electronic device according to an embodiment.

Referring to FIG. **5A**, an electronic device (e.g., the electronic device **101** of FIG. **1**, the electronic device **200** of FIG. **2A**, or the electronic device **400** of FIG. **4A**) may include a display **430** and a support plate **440** disposed on one surface of the display **430** facing the inside of the electronic device **400** (e.g., in +z direction). The display **430** may include a planar region **430a** and a deformable region **430b**.

According to an embodiment, the support plate **440** disposed on the planar region **430a** may be configured differently from the support plate **440** disposed on the deformable region **430b**. The planar region **430a** of the support plate **440** may extend from the deformable region **430b** of the support plate **440** in the first direction (e.g., in +y direction). For example, the support plate **440** may include a plurality of rigid regions (e.g., a first rigid region **440a'** and a second rigid region **440a''**) and a plurality of flexible regions (e.g., a first flexible region **440b'** and a second flexible region **440b''**) disposed between the plurality of rigid regions. The material of the support plate **440** disposed on the planar region **430a** may be different from that of the support plate **440** disposed on the deformable region **430b**. Due the difference in the materials, the physical properties (e.g., rigidity) of the support plate **440** disposed on the planar region **430a** and the support plate **440** disposed on the deformable region **430b** may be different from each other. For example, the support plate **440** disposed on the planar region **430a** may be formed of a material having higher rigidity than the support plate **440** disposed on the deformable region **430b**. For example, the plurality of flexible regions disposed in the planar region **430a** may have fewer openings than the plurality of flexible regions disposed in the deformable region **430b**. For example, an area of the plurality of rigid regions disposed in the planar region **430a** may be larger than an area of the plurality of rigid regions disposed in the deformable region **430b**. For example, the area of the support plate **440** disposed on the planar region **430a** and the support plate **440** disposed on the deformable region **430b** may be the same. The support plate **440** disposed on the planar region **430a** may further include at least one more rigid region than the support plate **440** disposed on the deformable region **430b**.

Referring to FIG. **5B**, unlike FIG. **5A**, the support plate **440** may include a planar plate **500** including a continuous surface in a planar region **430a**. The planar plate **500** including the continuous surface may have higher rigidity than the rigid regions. The support plate **440** may include a plurality of flexible regions (e.g., the first flexible region **440b'** and the second flexible region **440b''**) disposed between a plurality of rigid regions (e.g., the first rigid region **440a'** and the second rigid region **440a''**) and the plurality of rigid regions, in the deformable region **430b**. For example, the support plate **440** may include, in the deform-

able region **430b**, rigid regions **440a'** and **440a''** and a first flexible area **440b'** disposed between the rigid regions **440a'** and **440a''**.

Referring back to FIGS. **5A** and **5B**, the support plate **440** disposed in the planar region **430a** may continuously extend from the support plate **440** disposed in the deformable region **430b**. The support plate **440** disposed on the planar region **430a** may be connected to the support plate **440** disposed on the deformable region **430b**. For example, the support plate **440** disposed on the planar region **430a** may be made of the same material as the support plate **440** disposed on the deformable region **430b**. The support plate **440** disposed on the planar region **430a** and the support plate **440** disposed on the deformable region **430b** may be connected with the same material. The support plate **440** may be continuous at a boundary area in between the planar region **430a** and the deformable region **430b**. For example, the support plate **440** disposed on the planar region **430a** and the support plate **440** disposed on the deformable region **430b** may be welded with a different material from the support plate **440** and connected to each other. For example, the support plate **440** disposed in the planar region **430a** and the support plate **440** disposed in the deformable region **430b** may be coupled to each other using a fastening structure (e.g., a hook structure or a splice joint structure).

According to the above-described embodiment, the support plate **440** disposed on the display **430** of the electronic device **400** may be configured differently from the deformable region **430b** in the planar region **430a**, thereby reinforcing the planar region **430a** exposed to the outside in the first state and providing flexibility to the deformed area **430b** rolled up in the first state.

FIG. **6A** illustrates a portion of an internal structure of an electronic device with a gear head being disposed, according to an embodiment.

Referring to FIG. **6A**, an electronic device (e.g., the electronic device **101** of FIG. **1**, the electronic device **200** of FIG. **2A**, or the electronic device **400** of FIG. **4A**) may include a display **430**, a support plate **440**, a battery **610**, at least one gear head **620** and a guide gear **630**. According to an embodiment, the support plate **440** may include rigid regions **440a'** and **440a''** and a first flexible region **440b'** disposed between the rigid regions **440a'** and **440a''**. According to an embodiment, the support plate **440** may include a first layer **441** and a second layer **442**. However, the disclosure is not limited thereto. For example, the support plate **440** may further include a third layer disposed on the second layer **442**.

According to an embodiment, the first flexible region **440b'** may include a plurality of openings spaced apart at a certain length in a third direction (e.g., +x direction) perpendicular to the first direction (e.g., +y direction). For example, the first flexible region **440b'** may include a plurality of first openings included in the first layer **441** and a plurality of second openings included in the second layer **442**, having a certain length in the third direction. Each of the plurality of first openings may be spaced apart from each other in the first direction and/or the third direction. Each of the plurality of second openings may be spaced apart from each other in the first direction and/or the third direction. For example, one first opening **441a** of the plurality of first openings may be spaced apart from one second opening **442a** of the plurality of second openings in the first direction (e.g., +y direction). A portion of the first opening **441a** may overlap the second opening **442a**, when viewing the display **430** from above. Each of the plurality of first openings may

partially overlap each of the plurality of second openings, when viewing the display **430** from above.

According to an embodiment, the first layer **441** may include a plurality of first openings extending in the third direction (e.g., +x direction) in the first flexible region **440b'**. The second layer **442** may be disposed on the first layer **441** and may include a plurality of second openings in the first flexible region **440b'**. For example, one surface of each of the plurality of first openings may be in contact with the display **430**, and the other surface facing the one surface of each of the plurality of first openings may be in contact with the second layer **442**. For example, one surface of each of the plurality of second openings may be in contact with the first layer **441**, and the other surface facing one surface of each of the plurality of second openings may be directed to the inside of the electronic device **400** (e.g. in +z direction).

According to an embodiment, the first layer **441** may further include one first bridge **441b** in contact with one end of one first opening **441a**, among a plurality of first bridges. The second layer **442** may further include one second bridge **442b** in contact with one end of one second opening **442a**, among a plurality of second bridges. According to an embodiment, the first layer **441** may include a plurality of first openings and a plurality of first bridges causing the plurality of first openings to be spaced apart from each other. The second layer **442** may include a plurality of second openings and a plurality of second bridges causing the plurality of second openings to be spaced apart from each other. For example, each of the plurality of first bridges may be disposed between the plurality of first openings along the first direction (e.g., +y direction) and/or the third direction (e.g., +x direction) to space the plurality of first openings apart from each other. Each of the plurality of second bridges may be disposed between the plurality of second openings along the first direction and/or the third direction to space the plurality of second openings apart from each other.

According to an embodiment, the battery **610** may be disposed in an internal space of the electronic device **400**. The battery **610** may be disposed on the support plate **440**. The battery **610** may supply electric power to various electronic components within the electronic device **400**. For example, the battery **610** may be configured to supply electric power to a driving module (e.g., the driving module **260** of FIG. 3A) and a motor (e.g., the motor **261** of FIG. 3A) in the electronic device **400**, so that the driving module and the motor can transmit power to a pinion gear (e.g., the pinion gear **262** of FIG. 3A), a rack gear (e.g., the rack gear **263** of FIG. 3A), or a guide gear **630**. The battery **610** may supply electric power to a processor (e.g., the processor **120** of FIG. 1) or the display **430** in the electronic device **400**.

According to an embodiment, the at least one gear head **620** may transmit power transmitted from the motor **261** to the guide gear **630**, to the display **430** and the support plate **440**. For example, the at least one gear head **620** may engage with gear teeth **631** of the guide gear **630**. The at least one gear head **620** may guide so that the display **430** move in the first direction (e.g., +y direction) or in the second direction (e.g., -y direction) opposite to the first direction. The at least one gear head **620** may be arranged in both peripheries of the support plate **440** extending in a direction parallel to the first direction (e.g., +y direction), within a plurality of rigid regions (e.g., the first rigid region **440a'** and the second rigid region **440a''**) of the support plate **440**. However, the disclosure is not limited thereto. For example, the support plate **440** may include a plurality of rigid regions (e.g., the first rigid regions **440a'** and the second rigid region **440a''**)

disposed on both peripheries of the support plate **440** extending in a direction parallel to the first direction. The support plate **440** may include a plurality of flexible regions disposed between the rigid regions to space each of the rigid regions apart from each other. Each of the plurality of flexible regions may have the same length in the first direction. Each of the gear heads included in a first set of gear heads **620a** may be disposed on both peripheries of the support plate **440**, corresponding to each of the plurality of rigid regions and extending in the direction parallel to the first direction at both peripheries of the support plate **440**. Each of the gear heads included in the first set of gear heads **620a** may be disposed to protrude toward the inside of the electronic device **400** (e.g., in +z direction). Since each of the plurality of rigid regions is spaced apart at the same interval by each of the plurality of flexible regions, each of the gear heads included in the first set of gear heads **620a** may be correspondingly spaced at the same interval. The gear teeth **631** of the guide gear **630** may be inserted into a space formed between the gear heads included in the first set of gear heads **620a**.

According to an embodiment, the electronic device **400** may further include a guide gear **630** engaged with at least one gear head **620** and guiding movement of the display **430**. For example, the guide gear **630** may receive power from the motor **261** to guide the display **430** to move either in the first direction (e.g., +y direction) or in the second direction (e.g., -y direction) opposite to the direction, engaging with at least one gear head **620**. The guide gear **630** may engage with the gear heads included in the first set of gear heads **620a** arranged along both peripheries of the support plate **440**. For example, the guide gear **630** may include a first guide gear **630a** and a second guide gear **630b**. Each of the guide gears **630a** and **630b** may be disposed on both peripheries of the support plate **440** extending in the direction parallel to the first direction (e.g., +y direction). The first guide gear **630a** may engage with a plurality of gear heads arranged along one periphery of both peripheries of the support plate **440**. The second guide gear **630b** may engage with a plurality of gear heads arranged along the one periphery and the other periphery of both peripheries of the support plate **440**.

According to an embodiment, the electronic device **400** may further include an adhesive member **640** disposed between the first layer **441** and the second layer **442** disposed on the first layer, in a region corresponding to a plurality of rigid regions (e.g., the first rigid region **440a'** and the second rigid region **440a''**) of the support plate **440**. However, the disclosure is not limited thereto. For example, the support plate **440** may include a plurality of rigid regions and a plurality of flexible regions disposed between the plurality of rigid regions. The adhesive member **640** may be disposed between the first layer **441** and the second layer **442**, in the region corresponding to the plurality of rigid regions. The adhesive member **640** may be configured to maintain the shape of the support plate **440**.

According to an embodiment, the support plate **440** may further include a third layer disposed on the second layer **442**. The third layer may be configured to be substantially the same as or similar to the second layer **442**. The adhesive member **640** may be disposed between the second layer **442** and the third layer in a region corresponding to the plurality of rigid regions. For example, the adhesive member **640** may include an adhesive tape. According to an embodiment, the adhesive member **640** may be omitted. In the regions corresponding to the plurality of rigid regions, the first layer **441** and the second layer **442** may be attached by means of a thermal compression process. The second layer **442** and

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the third layer may be attached by means of a thermal compression process in the regions corresponding to the plurality of rigid regions. The support plate 440 may have a designated thickness in order to secure the rigidity of the rigid region 440a. The support plate 440 may be arranged with various shapes of a plurality of openings and slits so as to secure the ductility in the flexible region including the first flexible region 440b'. The number of layers of the support plate 440 may be determined depending on the shape and arrangement of the openings and slits.

According to the above-described embodiment, the support plate 440 can increase the rigidity to support the display 430 by means of attachment of the first layer 441 and the second layer 442 in the regions corresponding to the plurality of rigid regions. The support plate 440 may include, in the regions corresponding to the plurality of flexible regions, a first layer 441 including a plurality of first openings and a plurality of first bridges to space each of the plurality of first openings apart from each other, and a second layer 442 disposed on the first layer 441 and including a plurality of second openings and a plurality of second bridges to space each of the plurality of second openings apart from each other, thereby increasing the rigidity to support the deformable area of the display 430 (e.g., the second area 230b of FIG. 3B and the deformable area 430b of FIG. 4A), so that the support plate 440 can be configured to be deformed into a shape corresponding to the shape of the deformable region 430b. The electronic device 400 may include at least one gear head disposed on each of the plurality of rigid regions, and a guide gear 630 engaged with the gear head, thereby guiding the display 430 in which the support plate 440 is disposed.

FIGS. 6B and 6C are front views of a display structure of an electronic device with a gear head being disposed therein according to an embodiment.

Referring to FIGS. 6B and 6C, an electronic device (e.g., the electronic device 101 of FIG. 1, the electronic device 200 of FIG. 2A, or the electronic device 400 of FIG. 4A) may include a display 430, a support plate 440 and at least one gear head 620. According to an embodiment, the support plate 440 may include a plurality of rigid regions (e.g., the first rigid region 440a' and a second rigid region 440a'') and a plurality of flexible regions (e.g., a first flexible region 440b' and a second flexible region 440b'') disposed between the plurality of rigid regions. According to an embodiment, the support plate 440 may include a first layer (e.g., the first layer 441 of FIG. 4A) and a second layer (e.g., the second layer 442 of FIG. 4A), but the disclosure is not limited thereto.

According to an embodiment, the at least one gear head 620 may be disposed in the plurality of rigid regions (e.g., the first rigid region 440a' and the second rigid region 440a'') of the support plate 440. A first set of gear heads 620a may be disposed on both peripheries of the support plate 440 extending in the direction parallel to the first direction (e.g., the +y direction). A second set of gear heads 620b may be spaced apart by a designated distance from one of both the peripheries toward the other periphery. The gear heads included in the second set of gear heads 620b may be arranged in the direction parallel to the first direction. For example, the support plate 440 may include the rigid regions 440a' and 440a'' disposed on both peripheries of the support plate 440 extending in the direction parallel to the first direction, and extending from one of both peripheries up to the other one of both peripheries. The second rigid region 440a'' may include some of the first set of gear heads 620a disposed on both the peripheries. The second rigid region

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440a'' may include some of the second set of gear heads 620b spaced apart by a designated distance from one edge in the second rigid region 440a'' toward the other periphery. Some of the second set of gear heads 620b may reduce deflection (or sagging) of the display 430. According to an embodiment, the second set of gear heads 620b may be disposed at the center of both the peripheries of the support plate 440 extending in the first direction. According to an embodiment, referring to FIG. 6C, the second set of gear heads 620b may be omitted.

According to the above-described embodiment, the support plate 440 may include a set of second gear heads 620b disposed in the center of both peripheries parallel to the first direction (e.g., +y direction), thereby reducing the deflection of the central portion of the display 430 toward the inside of the electronic device 400 (e.g., in +z direction). Since the support plate 440 includes a plurality of layers, the rigidity for supporting the display 430 can be reinforced, so that some or all of the second set of gear heads 620b disposed in the center portion may be omitted. Such omission of some or all of the second set of gear heads 620b makes it possible to provide for an additional space for the battery 610 inside the electronic device 400.

FIG. 7A is an exploded perspective view of a display support structure of an electronic device according to an embodiment. FIGS. 7B and 7C are front views of a display support structure according to an embodiment.

Referring to FIGS. 7A, 7B, and 7C, the support plate 440 may include rigid regions 440a' and 440a'' (e.g., the first rigid region 440a' and the second rigid region 440a'' of FIG. 6B) and flexible regions 440b' and 440b'' (e.g., the first flexible region 440b' and the second flexible region 440b'' of FIG. 6B) disposed between the rigid regions 440a' and 440a''. The support plate 440 may include a first layer 441 and a second layer 442. The first layer 441 may include a plurality of first openings 441a, 741a, 741c, and 741e extending in a third direction (e.g., +x direction) in the first flexible region 440b'. The first layer 441 may include a plurality of first bridges 441b, 741b, and 741d to space apart the first openings 441a, 741a, 741c, and 741e from each other. The second layer 442 may be disposed on the first layer 441 and include a plurality of second openings 442a, 742a, 742c, and 742e. The second layer 442 may include a plurality of second bridges 442b, 742b, and 742d to space apart the second openings 442a, 742a, 742c, and 742e.

According to an embodiment, the first layer 441 may include one first opening 441a of a plurality of first openings 441a, 741a, 741c and 741e having a certain length in the third direction (e.g., +x direction) in the first flexible region 440b'. The first layer 441 may include one first bridge 441b in contact with one end 441a' of the first opening 441a and another first bridge 741b in contact with the other end 441a'' of the first opening 441a. The first layer 441 may include a first opening 741a disposed in a fourth direction (e.g., -x direction) opposite to the third direction (e.g., +x direction) with respect to the first opening 441a, and another first opening 741c disposed in the third direction with respect to the first opening 441a. The first opening 441a and the first opening 741a disposed in the fourth direction with respect to the one first opening 441a may be spaced apart by the one first bridge 441b. The first opening 441a and the first opening 741c disposed in the third direction with respect to the first opening 441a may be spaced apart by the other first bridge 741b. According to an embodiment, the first layer 441 may include a plurality of first intermediate bridges extending in the direction parallel to the third direction. For example, the first layer 441 may include a first opening 741e

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disposed in the first direction (e.g., +y direction) with respect to the one first opening 441a. The one first opening 441a and the first opening 741e disposed in the first direction with respect to the first opening 441a may be spaced apart by one first intermediate bridge 741d of a plurality of first intermediate bridges. The first layer 441 may include a first openings 741e disposed in the first direction with respect to the first opening 441a and a plurality of first openings spaced apart in the third direction and/or the fourth direction. However, the disclosure is not limited thereto, and the first layer 441 may include a plurality of first openings having a certain length in the third direction and spaced apart from each other in the first direction or the third direction. The first layer 441 may include a plurality of first intermediate bridges to space each of the plurality of first openings apart from each other in the direction parallel to the first direction.

According to an embodiment, the second layer 442 may be disposed on the first layer 441 and include a plurality of second openings 442a, 742a, 742c, and 742e in the first flexible region 440b'. The second layer 442 may have a configuration that is substantially the same as or similar to that of the first layer 441. For example, the second layer 442 may include one second opening 442a among the second openings 442a, 742a, 742c, and 742e. The second layer 442 may include a second bridge 442b in contact with one end 442a' of the second opening 442a and a second bridge 742b in contact with the other end 442a" of the second opening 442a. The second layer 442 may include a second opening 742c disposed in the third direction (e.g., +x direction) with respect to the second opening 442a and a second opening 742a disposed in the fourth direction (e.g., -x direction) opposite to the third direction with respect to the second opening 442a. The second layer 442 may include a second opening 742e disposed in the first direction (e.g., +y direction) with respect to the one second opening 442a and a plurality of second intermediate bridges extending in the direction parallel to the third direction. One second bridge 742d of the plurality of second intermediate bridges may cause the second opening 442a and a second opening 742e disposed in the first direction with respect to the second opening 442a to be spaced apart. The second layer 442 may further include a plurality of second openings 742e spaced apart in the third direction and/or the fourth direction from the second opening 742e disposed in the first direction with respect to the second opening 442a. The second layer 442 may include a plurality of second openings spaced apart from each other in the first direction and/or the third direction. The second layer 442 may include a plurality of second intermediate bridges to space each of the plurality of second openings apart from each other in the direction parallel to the first direction.

Referring to FIGS. 7A and 7B, the second layer 442 may include a plurality of second openings 442a, 742a, 742c, and 742e partially overlapping each of the first openings 441a, 741a, 741c and 741e included in the first layer 441 in the first flexible region 440b'. Referring to FIG. 7B, in the first flexible region 440b', a first bridge 441b in contact with one end 441a' of one first openings 441a among the first openings 441a, 741a, 741c, and 741e disposed in the first layer 441 may partially overlap one second opening 442a of the second openings 442a, 742a, 742c, and 742e disposed in the second layer 442. A part of the first bridge 441b may overlap one second bridge 442b in contact with one end 442a' of the second opening 442a. The remaining part of the first bridge 441b may be exposed through the second opening 442a, when the second layer 442 is viewed from above. Since one end 442a' of the second opening 442a is in contact with the

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rest of the first bridge 441b, the first opening 441a may partially overlap the second opening 442a. For example, in the first flexible region 440b', another first bridge 741b in contact with the other end 441a" of the first opening 441a may be partially overlapped with another second bridge 742b in contact with the other end 442a" of the second opening 442a. Since the other end 441a" of the first opening 441a overlaps the other second bridge 742b, the first opening 441a may partially overlap the second opening 442a. Such an overlapping portion of the first bridge 441b and the second bridge 442b can reinforce the rigidity of the first layer 441 and the second layer 442, thereby reducing the deflection (or sagging) of the surface of the support plate 440. Due to the overlap between the bridges 441b and 442b disposed on different layers, an area of the overlapping portion of the openings of the different layers can increase. Owing to increasing the overlapping area of the openings, it is possible to secure necessary ductility in the deformable region (e.g., the second region 230b of FIG. 3B and the deformable region 430b of FIG. 4A) of the support plate 440.

Referring back to FIGS. 7A and 7C, in the first flexible region 440b', one first bridge 441b in contact with one end 441a' of one first opening 441a may be in contact with one end 442a' of one second opening 442a, when viewing the second layer 442 from above, and all of the first bridge 441b may be exposed. One end 441a' of the first opening 441a is in contact with the first bridge 441b, and is exposed by the second opening 442a when the second layer 442 is viewed from above, and thus, the first opening 441a may partially overlap the second opening 442a.

According to an embodiment, in the first flexible region 440b', the other first bridge 741b in contact with the other end 441a" of the first opening 441a may come into contact with another second bridge 742b in contact with the other end 442a" of the second opening 442a, when the second layer 442 is viewed from above. When the second layer 442 is viewed from above, the entirety of the other first bridge 741b may be exposed by the second opening 742c disposed in the third direction (e.g., +x direction) with respect to the one second opening 442a. The first opening 441a may overlap the other second bridge 742b. The second opening 442a may overlap one first bridge 441b in contact with one end 441a' of the first opening 441a. A distance d4 from one end 441a' of the first opening 441a to the other second bridge 742b may be different from a distance d3 that is the length of the first opening 441a. For example, the distance d3 from one end 441a' of the first opening 441a to the other end 441a" of the first opening 441a may be different from the distance d4 from the one end 441a' of the first opening 441a to the other end 442a" of the second opening 442a close to the other end 441a" of the first opening 441a, amongst both ends of one second opening 442a of a plurality of second openings overlapping the first opening 441a. The distance d3 from one end 441a' of the first opening 441a to the other end 441a" of the first opening 441a may correspond to the length in the third direction (e.g., the +x direction) of the first opening 441a. The distance d4 from the one end 441a' of the first opening 441a to the other end 442a" of the second opening 442a close to the other end 441a" of the first opening 441a, amongst both ends of one second opening 442a of a plurality of second openings overlapping the first opening 441a, may correspond to the distance from one end 441a' of the first opening 441a to another second bridge 742b overlapping the first opening

441a. When the other second bridge 742b is disposed to overlap the first opening 441a, the distance d4 may be shorter than the distance d3.

According to an embodiment, a length of each of the plurality of first openings in the third direction (e.g., +x direction) may be substantially the same as that of each of the plurality of second openings in the third direction. For example, one first opening 441a of the first openings 441a, 741a, 741c, and 741e may overlap one second opening 442a of the second openings 442a, 742a, 742c, and 742e. The distance d3 from one end 441a' of the first opening 441a to the other end 441a'' of the first opening 441a may be substantially the same as a distance d5 from one end 442a' of the second opening 442a to the other end 442a'' of the second opening 442a. For example, the length of each of the first openings 441a, 741a, 741c, and 741e in the third direction (e.g., +x direction) may be substantially equal to each other as the distance d3. The length of each of the second openings 442a, 742a, 742c, and 742e in the third direction may be substantially equal to each other as the distance d5. The length of each of the first openings 441a, 741a, 741c, and 741e in the third direction may be substantially equal to the length of each of the second openings 442a, 742a, 742c, and 742e in the third direction. However, the disclosure is not limited thereto, and the length of each of the plurality of first openings included in the plurality of flexible regions in the third direction may be different from that of each of the plurality of second openings included in the plurality of flexible regions in the third direction.

According to an embodiment, the width of each of the plurality of first openings in the first direction (e.g., +y direction) perpendicular to the third direction (e.g., +x direction) may be substantially the same as the width in the first direction of each of the plurality of second openings. For example, the width of one first opening 441a in the first direction (e.g., +y direction) may correspond to the width in the first direction of one first bridge 441b in contact with the first opening 441a. The width of one second opening 442a in the first direction may correspond to the width in the first direction of one second bridge 442b in contact with the second opening 442a. The width of the first bridge 441b in the first direction may be substantially the same as the width of the second bridge 442b in the first direction. The width of the first opening 441a in the first direction may be substantially the same as the width of the second opening 442a in the first direction. The disclosure is not limited thereto, and the width in the first direction of each of the plurality of first bridges in contact with each of the plurality of first openings may be substantially the same as the width in the first direction of each of the plurality of second bridges in contact with each of the plurality of second openings. Since the width of each of the plurality of first bridges in the first direction is the same as the width of each of the plurality of second bridges in the first direction, the width of each of the plurality of first openings in the first direction may be substantially the same as the width in the first direction of each of the plurality of second openings.

According to the above-described embodiment, the support plate 440 may include a plurality of first openings (e.g., the first openings 441a, 741a, 741c, and 741e) and a plurality of second openings (e.g., the second openings 442a, 742a, 742c, and 742e) partially overlapping each of the plurality of first openings, in a plurality of flexible regions (e.g., the first flexible regions 440b' and the second flexible regions 440b''), thereby reinforcing the rigidity to support the deformable region 430b of the display 430 supported by the plurality of flexible regions. A plurality of

first bridges (e.g., the first bridges 441b, 741b, and 741d) to space apart each of the plurality of first openings may partially overlap a plurality of second bridges (e.g., the second bridges 442b, 742b, and 742d) to space apart each of the plurality of second openings, thereby reinforcing the supporting rigidity of the deformable region 430b.

FIGS. 7D and 7E are front views of a display support structure of an electronic device according to an embodiment.

Referring to FIG. 7D, the support plate 440 may include rigid regions 440a' and 440a'' and flexible regions 440b' and 440b'' (e.g., the first flexible region 440b' and the second flexible region 440b'' of FIG. 6B) disposed between the rigid regions 440a' and 440a''. The support plate 440 may include a first layer (e.g., the first layer 441 of FIG. 7A) and a second layer 442 (e.g., the second layer 442 of FIG. 7A). The first layer 441 may include first openings (e.g., the first openings 441a, 741a, 741c and 741e of FIG. 7A) extending in the third direction (e.g., +x direction) in the first flexible region 440b'. The second layer 442 may be disposed on the first layer 441 and include second openings 442a, 742a, 742c, and 742e. The first layer 441 may include first bridges (e.g., the first bridges 441b, 741b, and 741d of FIG. 7A) to get the first openings 441a, 741a, 741c, and 741e to be spaced apart from each other. The second layer 442 may include second bridges 442b, 742b, and 742d to get the second openings 442a, 742a, 742c, and 742e to be spaced apart from each other.

According to an embodiment, the second layer 442 may include a plurality of second openings 442a, 742a, 742c, and 742e partially overlapping each of the first openings 441a, 741a, 741c, and 741e included in the first layer 441, in the first flexible region 440b'. For example, in the first flexible region 440b', the first layer 441 may include one first opening (e.g., one first opening 441a of FIG. 7A) of the first openings 441a, 741a, 741c, and 741e. When the second layer 442 is viewed from above, one first bridge 441b in contact with one end 441a' of the first opening 441a may be exposed by one second opening 442a of the second openings 442a, 742a, 742c, and 742e.

According to an embodiment, a distance s1 from one end 442a' of the one second opening 442a to the first bridge 441b may be substantially the same as a distance s2 from the other end 442a'' of the second opening 442a to the first bridge 441b. For example, when the second layer 442 is viewed from above, one end of one first bridge 441b may face one end 442a' of one second opening 442a, and the other end of the first bridge 441b may be directed to the other end 442a'' of the second opening 442a. The other end of the first bridge 441b may be in contact with one end 441a' of one first opening 441a.

Heretofore, although it has been described that the distance from both ends of the second opening 442a to the first bridge 441b is substantially the same, the disclosure is not limited thereto, and the distance s1 and the distance s2 may be different from each other.

Referring to FIG. 7E, the support plate 440 may further include a third layer 743. The third layer 743 may be disposed on the second layer 442 and may include third openings 743a, 743c, and 743e in the first flexible region 440b'. The third layer 743 may include third bridges 743b and 743d to space apart each of the third openings 743a, 743c, and 743e. The third layer 743 may have a configuration substantially the same as or similar to that of the second layer 442. For example, the third layer 743 may include one third opening 743a of the third openings 743a, 743c, and 743e and one third bridge 743b in contact with one end 743a'

of the one third opening 743a. The third layer 743 may further include a third opening 743c disposed in the third direction (e.g., +x direction) with respect to the one third opening 743a. The one third opening 743a and the third opening 743c disposed in the third direction with respect to the one third opening 743a may be spaced apart by the one third bridge 743b. The third layer 743 may include a plurality of third intermediate bridges extending in the third direction and a third opening 743e disposed in the first direction (e.g., +y direction) with respect to the one third opening 743a. The one third opening 743a and the third opening 743e disposed in the first direction (e.g., +y direction) with respect to the one third opening 743a may be spaced apart by a third intermediate bridge 743d of one of the plurality of third intermediate bridges. The disclosure being not limited thereto, the third layer 743 may include a plurality of third openings respectively spaced apart from each other in the direction parallel to the first direction and/or in the direction parallel to the third direction. The third layer 743 may include a plurality of third intermediate bridges to space the plurality of third openings apart from each other in the direction parallel to the first direction.

According to an embodiment, one third opening 743a of the third openings 743a, 743c, and 743e included in the third layer 743 may partially overlap one first opening 441a of the first openings 441a, 741a, 741c, and 741e facing the third direction (e.g., the +x direction), in the first flexible region 440b'. The third opening 743a may partially overlap one second opening 442a of the second openings 442a, 742a, 742c, and 742e. The one third opening 743a may overlap one first bridge 441b in contact with one end 441a' of the first opening 441a and one second bridge 442b in contact with one end 442a' of the second opening 442a. The distance d3 from one end 441a' of the first opening 441a to the other end 441a'' of the first opening 441a may be different from a distance d6 from one end 441a' of the first opening 441a to one end of the third opening close to the other end 441a'' of the first opening 441a, amongst both ends of one third opening of the third openings 743a, 743c, and 743e overlapping the first opening 441a. For example, one first opening 441a of the first openings 441a, 741a, 741c, and 741e may respectively partially overlap one third opening 743a of the third openings 743a, 743c, and 743e and a third opening 743c disposed in the third direction (e.g., +x direction) with respect to the one third opening 743a. One end 441a' of the first opening 441a may be exposed by the one third opening 743a, when the third layer 743 is viewed from above. The other end 441a'' of the first opening 441a may be exposed by the third opening 743c disposed in the third direction (e.g., +x direction) with respect to the one third opening 743a, when the third layer 743 is viewed from above. Among the third openings 743a, 743c, and 743e, one end 743c' of the third opening 743c disposed in the third direction (e.g., +x direction) with respect to the one third opening 743a may be an end close to the other end 441a'' of the first opening 441a. The distance d3 from one end 441a' of the first opening 441a to the other end 441a'' of the first opening 441a may be different from a distance d6 from one end 441a' of the first opening 441a to one end 743c' of the third opening 743c disposed in the third direction (e.g., +x direction) with respect to the one third opening 743a.

According to an embodiment, the third layer 743 may include one third bridge 743b in contact with one end 743a' of one third opening 743a. The one third bridge 743b may overlap one first opening 441a and one second opening 442a. The first layer 441 may include one first bridge 441b in contact with one end 441a' of the first opening 441a, and

another first bridge 741b in contact with the other end 441a'' of the first opening 441a. The second layer 442 may include one second bridge 442b in contact with one end 442a' of the one second opening 442a. According to an embodiment, the distance s3 from the first bridge 441b to the second bridge 442b may be substantially the same as the distance s4 from the second bridge 442b to the one third bridge 743b. The distance s3 from the first bridge 441b to the second bridge 442b may be substantially the same as the distance s5 from the one third bridge 743b to the other first bridge 741b. The distance s5 from the one third bridge 743b to the other first bridge 741b may be a distance s5 to the other end 441a'' of the first opening 441a facing one end 441a' of the first opening 441a. For example, a first bridge 441b in contact with one end 441a' of a first opening 441a and a second bridge 442b in contact with one end 442a' of a second opening 442a may be exposed through a third opening 743a, when the third layer 743 is viewed from above. Another first bridge 741b in contact with the other end 441a'' of the first opening 441a and another second bridge 742b in contact with the other end 442a'' of the second opening 442a may be exposed through a third opening 743c disposed in the third direction (e.g., +x direction) with respect to the one third opening 743a, when the third layer 743 is viewed from above. The third opening 743a, and the third opening 743c disposed in the third direction (e.g., +x direction) with respect to the third opening 743a may be spaced apart by the third bridge 743b.

Heretofore, although it has been described that when the third layer 743 is viewed from above, the bridges 441b, 442b, 743b, and 741b are spaced apart at equal intervals, the disclosure is not limited thereto, and the distance s3, distance s4 and the distance s5 may be different from each other.

According to an embodiment, in regions corresponding to the rigid regions 440a' and 440a'', an adhesive member (e.g., the adhesive member 640 of FIG. 6A) may be disposed in between the second layer 442 and the third layer 743. For example, the adhesive member 640 may be disposed between the first layer 441 and the second layer 442 in the region corresponding to the rigid regions 440a' and 440a''. The adhesive member 640 may be disposed between the second layer 442 and the third layer 743 in the region corresponding to the rigid regions 440a' and 440a'' to connect the first layer 441, the second layer 442, and the third layer 743 in the support plate 440.

According to the above-described embodiment, in a plurality of flexible regions (e.g., the first flexible region 440b' and the second flexible region 440b''), the support plate 440 may include a plurality of first openings (e.g., first openings 441a, 741a, 741c, and 741e), a plurality of second openings (e.g., the second openings 442a, 742a, 742c, and 742e) partially overlapping each of the plurality of first openings, and a plurality of third openings (e.g., the third openings 743a, 743c, and 743e) partially overlapping each of the plurality of first openings or the plurality of second openings, thereby reinforcing the supporting rigidity of the deformable region (e.g., the second region 230b of FIG. 3B or the deformable region 430b of FIG. 4A) of the display (e.g., the display module 160 of FIG. 1, the display 230 of FIG. 2A, or the display 430 of FIG. 4A) supported by the plurality of flexible regions. A plurality of bridges included in each of the different layers (e.g., first bridges 441b, 741b and 741d, second bridges 442b, 742b and 742d, and third bridges 743b and 743d) may be arranged being spaced apart

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from each other, so that the rigidity of the support plate **440** can be reinforced while securing the ductility of the deformation region **430b**.

FIG. **8A** is a front view of a display support structure of an electronic device according to an embodiment. FIG. **8B** is a cross-sectional view illustrating an example of a display support structure of an electronic device taken along a line B-B' of FIG. **8A** according to an embodiment. FIG. **9A** is a front view of a display support structure of an electronic device according to an embodiment. And FIG. **9B** is a cross-sectional view illustrating an example of a display support structure of an electronic device taken along a line C-C' of FIG. **9A** according to an embodiment.

Referring to FIGS. **8A**, **8B**, **9A** and **9B**, the support plate **440** may include the rigid regions **440a'** and **440a''** and flexible regions (e.g., the first flexible region **440b'** and the second flexible region **440b''** of FIG. **6B**) disposed between the rigid regions **440a'** and **440a''**. The support plate **440** may include a first layer **441** and a second layer **442**. The first layer **441** may include a plurality of first openings extending in the third direction (e.g., +x direction) in the first flexible region **440b'**. The first layer **441** may include a plurality of first bridges to space the plurality of first openings apart from each other. The second layer **442** may be disposed on the first layer **441** and include a plurality of second openings. The second layer **442** may include second bridges to space the plurality of second openings apart. The first bridges **441b**, **741b** and **741d**, the second openings **442a**, **742a**, **742c** and **742e**, and the second bridges **442b**, **742b** and **742d** may respectively correspond to the first bridges **441b**, **741b** and **741d**, the second openings **442a**, **742a**, **742c** and **742e**, and the second bridges **442b**, **742b** and **742d** of FIGS. **7A** to **7E**.

According to an embodiment, the second layer **442** may include a plurality of second openings each fully overlapping each of a plurality of first openings included in the first layer **441** in the first flexible region **440b'**. For example, referring to FIGS. **8A** and **8B**, in a region corresponding to the first flexible region **440b'**, the second layer **442** may include a plurality of second bridges disposed on both peripheries extending in the direction parallel to the first direction (e.g., +y direction). One second bridge **442b** of the plurality of second bridges may be in contact with one end **442a'** of one second opening **442a**, and another first bridge **741b** may be in contact with the other end **442a''** of the second opening **442a**. The second bridge **442b** and the other second bridge **742b** may be respectively disposed on both peripheries of the second layer **442** extending in the direction parallel to the first direction. When the second layer **442** is viewed from above, one first bridge **441b** in contact with one end **441a'** of one first opening **441a** and the other first bridge **741b** in contact with the other end **441a''** of the first opening **441a** may be entirely exposed through the second opening **442b**. The first opening **441a** may entirely overlap the second opening **442a**. The disclosure is not limited thereto, and each of the plurality of first openings may entirely overlap each of the plurality of second openings in the region corresponding to the plurality of flexible regions.

According to an embodiment, the second layer **442** may include a plurality of second openings partially overlapping each of the plurality of first openings included in the first layer **441**, in the first flexible region **440b'**. For example, referring to FIGS. **9A** and **9B**, a plurality of second intermediate bridges (e.g., a second intermediate bridge **742d** in FIG. **7A**) in a region corresponding to the first flexible region **440b'** may be omitted. The plurality of second bridges in the first flexible region **440b'** may each extend in the

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direction parallel to the first direction (e.g., +y direction) to connect the rigid regions **440a'** and **440a''**. A portion of one first opening **441a** among the plurality of first openings may overlap one second opening **442a**. The remaining portion of the first opening **441a** may overlap the second opening **742c** disposed in the third direction (e.g., +x direction) with respect to the second opening **442a**. For example, each of the plurality of first bridges may entirely overlap the plurality of second bridges. Since each of the plurality of first bridges is in contact with one end of each of the plurality of first openings, each of the plurality of first openings may partially overlap each of the plurality of second bridges spaced apart by the plurality of second bridges.

According to an embodiment, one end **441a'** of one first opening of the plurality of first openings **441a** may be closer to one end **442a'** of both ends of one second opening **442a** partially overlapping the first opening **441a**. The other end **441a''** of the first opening **441a** may be closer to the other end **442a''** of both ends of the second opening **442a**. The distance **d3** from one end **441a'** of the first opening **441a** to the other end **441a''** of the first opening **441a** may be different from the distance **d4** from the one end **441a'** of the first opening **441a** to the other end **442a''** of the second opening **442a** close to the other end **441a''** of the first opening **441a**, amongst both ends of one second opening **442a** of a plurality of second openings overlapping the first opening **441a**. For example, referring to FIGS. **8A** and **8B**, one end **441a'** of one first opening **441a** may be close to one end **442a'** of both ends of one second opening **442a**, and the other end **441a''** of the first opening **441a** may be close to the other end **442a''** of both ends of the second opening **442a**. The distance **d3** from one end **441a'** of the first opening **441a** to the other end **441a''** of the first opening **441a** corresponds to a length in the third direction (e.g., +x direction) of the first opening **441a**, and thus, the distance **d3** may be different from the distance **d4** from one end **441a'** of the first opening **441a** to the other end **442a''** of the second opening **442a** close to the other end **441a''** of the first opening **441a**. For example, referring to FIGS. **9A** and **9B**, one end **441a'** of the first opening **441a** may be close to one end **442a'** of one second opening **442a**, and the other end **441a''** of the first opening **441a** may be close to one end of the second opening **742c** disposed in the third direction (e.g., +x direction) with respect to the second opening **442a**. Since the second opening **442a** and the second opening **742c** disposed in the third direction with respect to the second opening **442a** are spaced apart from each other in the third direction, the distance **d3** from one end **441a'** of the first opening **441a** to the other end **441a''** of the first opening **441a** may be different from the distance **d4** from one end **441a'** of the first opening **441a** to one end of the second opening **742c** disposed in the third direction with respect to the second opening **442a**, close to the other end **441a''** of the first opening **441a**.

According to an embodiment, the first layer **441** may include a first bridge **441b** in contact with one end **441a'** of a first opening **441a**, and the second layer **442** may include a second bridge **442b** in contact with one end **442a'** of a second opening **442a**. The distance **d3** from the other end **441a''** of the first opening **441a** to the first bridge **441b** may be different from the distance **d5** from the other end **442a''** of the second opening **442a** to the second bridge **442b**. For example, referring to FIGS. **8A** and **8B**, a first bridge **441b** in contact with one end **441a'** of a first opening **441a** and another first bridge **741b** in contact with the other end **441a''** of the first opening **441a** may be fully exposed through the second opening **442a**. Since both ends of the first opening

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441a are fully exposed through the second opening 442a, the distance d3 from the other end 441a" of the first opening 441a to the first bridge 441b may be different from the distance d5 from the other end 442a" of the second opening 442a to the second bridge 442b. For example, referring to FIGS. 9A and 9B, a second opening 442a and a second opening 742c disposed in the third direction (e.g., +x direction) with respect to the second opening 442a may partially overlap the first opening 441a, respectively. One end 441a' of the first opening 441a is close to one end 442a' of the second opening 442a, and the other end 441a" of the first opening 441a is close to one end of the second opening 742c disposed in the third direction with respect to the second opening 442a, so the distance d3 from the other end 441a" of the first opening 441a to the first bridge 441b may be different from the distance d5 from the other end 442a" of the second opening 442a to the second bridge 442b.

According to the above-described embodiment, in a plurality of flexible regions (e.g., the first flexible region 440b' and the second flexible region 440b"), the support plate 440 includes a plurality of first openings (e.g., the first openings 441a, 741a, 741c, and 741e) and a plurality of second openings (e.g., the second openings 442a, 742a, 742c, 742e) partially and/or fully overlapping each of the plurality of first openings, so that the display (e.g., the display module 160 of FIG. 1, the display 230 of FIG. 2A, or the display 430 of FIG. 4A) supported by the plurality of flexible regions can be reinforced with more rigidity to support the deformable region (e.g., the second region 230b of FIG. 3B, or the deformable region 430b of FIG. 4A). The plurality of first bridges (e.g., the first bridges 441b, 741b, and 741d) to space the plurality of first openings apart from each other can be partially and/or fully overlapped with the plurality of bridges (e.g., the second bridges 442b, 742b, and 742d) spacing apart the plurality of second openings from each other, thereby reinforcing the supporting rigidity for the deformable region 430b.

FIG. 10 is a side view of a display support structure of an electronic device according to an embodiment.

Referring to FIG. 10, the support plate 440 may include rigid regions 440a' and 440a" and flexible regions (e.g., the first flexible region 440b' and the second flexible region 440b") of FIG. 6B) disposed between the rigid regions. The support plate 440 may include a first layer 441, a second layer 442, and a third layer 743. The first layer 441 may include a plurality of first openings (e.g., the first openings 441a, 741a, 741c, and 741e of FIG. 7A) extending in the third direction (e.g., +x direction) in the first flexible region 440b'. The second layer 442 may be disposed on the first layer 441 and include a plurality of second openings (e.g., the second openings 442a, 742a, 742c, and 742e of FIG. 7A). The third layer 743 may be disposed on the second layer 442 and include a plurality of third openings (e.g., the third openings 743a, 743c, and 743e of FIG. 7E). The first layer 441 may include a plurality of first bridges (e.g., the first bridges 441b, 741b, and 741d of FIG. 7A) spacing the plurality of first openings apart. The second layer 442 may include a plurality of second bridges (e.g., the second bridges 442b, 742b, and 742d of FIG. 7A) spacing the plurality of second openings apart. The third layer 743 may include third bridges (e.g., the third bridges 743b and 743d of FIG. 7E) spacing the plurality of third openings apart. The disclosure is not limited thereto, and the support plate 440 may include a plurality of overlapping layers, when the support plate 440 is viewed from above.

According to an embodiment, the thickness d7 of the first flexible region 440b' may be less than or equal to the

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thickness d8 of the rigid regions 440a' and 440a". For example, in the first flexible region 440b', the first layer 441 may include a first bridge 441b in contact with a first opening 441a, the second layer 442 may include a second bridge 442b in contact with a second opening 442a, and the third layer 743 may include a third bridge 743b in contact with a third opening 743a. The thickness of the first layer 441 included in the rigid regions 440a' and 440a" may be substantially the same as the thickness of the first bridge 441b or may be thicker than the thickness of the first bridge 441b. The thickness of the second layer 442 included in the rigid regions 440a' and 440a" may be substantially the same as the thickness of the second bridge 442b or may be thicker than the thickness of the second bridge 442b. The thickness of the third layer 743 included in the rigid regions 440a' and 440a" may be substantially the same as the thickness of the third bridge 743b or may be thicker than the thickness of the third bridge 743b. The sum of the thicknesses of the bridges 441b, 442b, and 743b may be substantially as the thickness d8 of the rigid regions 440a' and 440a", or may be thinner than the thickness d8 of the rigid regions 440a' and 440a". The thickness of the support plate 440 included in the plurality of rigid regions may be substantially equal to or greater than the thickness of each of the plurality of flexible regions disposed between the plurality of rigid regions.

According to the above-described embodiment, the support plate 440 may be disposed between a plurality of rigid regions (e.g., the first rigid region 440a' and the second rigid region 440a"), and include a plurality of flexible regions (e.g., the first flexible regions 440b' and the second flexible regions 440b") thinner than the plurality of rigid regions. The support plate 440 can reinforce, through the plurality of layers 441, 442, and 743, the rigidity of the deformable region (e.g., the second region 230b of FIG. 3B and the deformable region 430b of FIG. 4A) of the display (e.g., the display module 160 of FIG. 1, the display 230 of FIG. 2A, or the display 430 of FIG. 4A). The support plate 440 can provide elasticity or flexibility when deformed into shape according to the deformation of the display 430, through the first flexible region 440b' having a thickness smaller than that of the rigid regions 440a' and 440a".

FIGS. 11A and 11B are side views illustrating a display structure of an electronic device according to an embodiment.

Referring to FIG. 11A, an electronic device (e.g., the electronic device 101 of FIG. 1, the electronic device 200 of FIG. 2A, or the electronic device 400 of FIG. 4A) may include a display 430, a support plate 440, at least one gear head 620, and at least one support bar 1100. The support plate 440 may include a plurality of rigid regions (e.g., the first rigid regions 440a' and the second rigid regions 440a") and a plurality of flexible regions (e.g., the first flexible region 440b' and the second flexible region 440b") disposed between the plurality of rigid regions. The support plate 440 may include a first layer 441, a second layer 442 and a first reinforcement layer 1110.

According to an embodiment, the at least one support bar 1100 may be configured to support the support plate 440. When the display 430 is viewed from above, the at least one support bar 1100 may overlap each of the plurality of rigid regions. One surface of the at least one support bar 1100 may be in contact with the support plate 440 included in the plurality of rigid regions, and the other surface facing the one surface of the at least one support bar 1100 may be in contact with at least one gear head 620. For example, the support plate 440 may include a first layer 441 in a direction toward the inside of the electronic device 400 and a second

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layer **442** disposed on the first layer **441**. At least one support bar **1100** may be disposed on one surface facing the inside of the electronic device **400** and corresponding to the rigid regions **440a'** and **440a''** of the second layer. At least one gear head **620** may be disposed on the other surface facing one surface of the at least one support bar **1100** in contact with the second layer.

According to an embodiment, the at least one gear head **620** may be disposed to overlap the at least one support bar **1100** when the display **430** is viewed from above. For example, each of the gear heads included in a first set of gear heads (e.g., a first set of gear heads **620a** of FIG. 6A) may be in contact with a plurality of support bars disposed on a plurality of rigid regions. Each of the gear heads may overlap the plurality of support bars, when the display **430** is viewed from above.

According to an embodiment, the support plate **440** may further include a first reinforcement layer **1110** disposed between the first layer **441** and the second layer **442**. The first reinforcement layer **1110** may include a continuous surface having a shape corresponding to one surface of the display **430** in a plurality of flexible regions. For example, in a region corresponding to the first flexible region **440b'**, the first layer **441** may include a first opening **441a** and a first bridge **441b** in contact with the first opening **441a**, and the second layer **442** may include a second opening **442a** and a second bridge **442b** in contact with the second opening **442a**. In a region corresponding to the first flexible region **440b'**, one surface of the first reinforcement layer **1110** may face the first opening **441a** and the first bridge **441b**, and the other surface facing the one surface of the first reinforcement layer **1110** may be directed to the second opening **442a** and the bridge **442b**. The one surface of the first reinforcement layer **1110** and the other surface facing the one surface may be continuous. The disclosure is not limited thereto, and in a region corresponding to the plurality of flexible regions of the support plate **440**, the first reinforcement layer **1110** may be formed such that one surface facing the first layer **441** and the other surface facing the one surface toward the second layer **442** are substantially continuous.

According to an embodiment, the first reinforcement layer **1110** may be in contact with the first layer **441** and the second layer **442** in a region corresponding to the plurality of rigid regions. For example, in the region corresponding to the plurality of rigid regions (e.g., the first rigid region **440a'** and the second rigid region **440a''**), an adhesive member (e.g., the adhesive member **640** of FIG. 6A) may be disposed between one surface facing the first layer of the first reinforcement layer **1110** and the first layer **441**. In the region corresponding to the plurality of rigid regions, an adhesive member **640** may be placed between another surface facing the one surface of the first reinforcement layer **1110** toward the second layer **442** and the first layer **441**.

According to an embodiment, the first reinforcement layer **1110** may reinforce the rigidity of the support plate **440** or compensate for thermal properties of the support plate **440**. For example, the first reinforcement layer **1110** may include a metal or a carbon fiber composite material to reinforce the rigidity of the support plate **440**. The first reinforcement layer **1110** may include any one of titanium, GFRP, CFRP, or plastic. The first reinforcement layer **1110** may include a metal material having high thermal conductivity to transfer heat emitted from the display **430**, in order to compensate for the thermal properties of the support plate **440**, or the display **430**, or a polymer material having low thermal conductivity, in order to reduce the heat transferred to the display **430**.

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According to an embodiment, the material of the first reinforcement layer **1110** may be different from that of the deformable region **430b** in a planar area of the display **430** (e.g., the first region **230a** in FIG. 3B or the planar region **430a** in FIG. 4A). For example, the material of the first reinforcement layer **1110** in the deformable region **430b** may have higher elasticity than the material of the first reinforcement layer **1110** in the planar region **430a**. For example, the material of the first reinforcement layer **1110** in the planar region **430a** may radiate more heat from the electronic device **400** than the material of the first reinforcement layer **1110** in the deformable region **430b**, or may insulate more heat from the outside of the electronic device **400**.

Heretofore, the description has been made on the basis of the first reinforcement layer **1110**, but the disclosure is not limited thereto. The support plate **440** may include a plurality of reinforcement layers disposed between the plurality of layers. In the reinforcement layers, the material of the planar region **430a** of the display **430** may be different from the material of the deformable region **430b** thereof. The plurality of reinforcement layers can be made of various materials, so that the support plate **440** can protect the display **430** from external impact or environmental alteration and further improve the functionality of the display **430**.

Referring to FIG. 11B, the support plate **440** may further include a third layer **743** and a second reinforcement layer **1120**. According to an embodiment, as opposed to FIG. 11A, the support plate **440** may not include the support bar **1100**. The third layer **743** and the second reinforcement layer **1120** may support the first layer **441**, the second layer **442**, the first reinforcement layer **1110** and the display **430**, in place of the support bar **1100**.

According to an embodiment, the at least one gear head **620** may be disposed on one surface of the third layer **743** facing the inside of the electronic device **400**, in an area corresponding to the plurality of rigid regions. For example, each of the gear heads included in a first set of gear heads (e.g., the first set of gear heads **620a** of FIG. 6A) may come into contact with one surface of a third layer **743** disposed on a plurality of rigid regions. Each of the gear heads may overlap each of the plurality of rigid regions, when the display **430** is viewed from above.

According to an embodiment, the second reinforcement layer **1120** may be disposed between the second layer **442** and the third layer **743**. The second reinforcement layer **1120** may be configured substantially the same as or similar to the first reinforcement layer **1110**. Similar to the first reinforcement layer **1110**, the second reinforcement layer **1120** may include a continuous surface having a shape corresponding to one surface of the display **430** in a plurality of flexible regions. For example, in a region corresponding to the first flexible region **440b'**, the second layer **442** may include a second opening **442a** and a second bridge **442b** in contact with the second opening **442a**, and the third layer **743** may include a third opening **743a** and a third bridge **743b** in contact with the third opening **743a**. In a region corresponding to the first flexible region **440b'**, one surface of the second reinforcement layer **1120** may face the second opening **442a** and the second bridge **442b**, and another surface facing the one surface of the second reinforcement layer **1120** may be directed to the third opening **743a** and the third bridge **743b**. The one surface of the second reinforcement layer **1120** and the other surface facing the one surface may be continuous. The disclosure is not limited thereto, and in regions corresponding to the plurality of flexible regions of the support plate **440**, the second reinforcement layer **1120** may be formed such that one surface facing the second layer

442 and the other surface facing the one surface toward the third layer **743** are continuous.

According to an embodiment, the second reinforcement layer **1120** may be in contact with the second layer **442** and the third layer **743**, in a region corresponding to the plurality of rigid regions. For example, in the region corresponding to the plurality of rigid regions (e.g., the first rigid region **440a'** and the second rigid region **440a''**), an adhesive member **640** may be disposed between one surface facing the second layer **442** of the second reinforcement layer **1120** and the second layer **442**. In the region corresponding to the plurality of rigid regions, an adhesive member **640** may be placed between the other surface facing the one surface of the second reinforcement layer **1120** and directing toward the third layer **743**, and the third layer **743**. According to the above-described embodiment, the electronic device **400** includes at least one support bar **1100** disposed in a plurality of rigid regions, thereby supporting at least one gear head **620** engaged with the guide gear (e.g., the guide gear **630** of FIG. **6A**) and supporting the display **430**. The support plate **440** can reinforce the rigidity to support the display **430** owing to having the plurality of reinforcement layers (e.g., the reinforcement layers **1110** and **1120**) disposed between the plurality of layers. Since the support plate **440** includes a plurality of layers and a plurality of reinforcement layers disposed between the plurality of layers, the at least one support bar **1100** may be omitted. Omitting the at least one support bar **1100** makes it possible to provide the electronic device **400** with a space for accommodating an extended battery (e.g., the battery **610** of FIG. **6A**). In a region corresponding to the plurality of rigid regions, more particularly, in the region where each of the plurality of layers is in contact with the reinforcement layers, corresponding to the plurality of flexible regions disposed between the plurality of rigid regions, each of the plurality of layers includes a continuous surface, so that the support plate **440** can increase the rigidity of supporting the deformable area (e.g., the second area **230b** of FIG. **3B** and the deformable area **430b** of FIG. **4A**) of the display **430**.

According to the above-described embodiment, an electronic device (e.g., an electronic device **101** of FIG. **1**, an electronic device **200** of FIG. **2A**, or an electronic device **400** of FIG. **4A**) may include a first housing (e.g., a first housing **210** of FIG. **2A** or a first housing **410** of FIG. **4A**), a second housing (e.g., a second housing **220** of FIG. **2A** or a second housing **420** of FIG. **4A**) coupled to the first housing so as to be slidable with respect to the first housing in a first direction (e.g., +y direction) and a second direction (e.g., +y direction) opposite to the first direction, a display (e.g., a display module **160** of FIG. **1**, a display **230** of FIG. **2A**, a display **430** of FIG. **4A**) including a planar region (e.g., a first area **230a** of FIG. **3B** or a planar region **430a** of FIG. **4A**) disposed on the second housing and a deformable region (e.g., a second region **230b** of FIG. **3B** or a deformable region **430b** of FIG. **4A**) that contacts the planar region and is exposed to an outside as the second housing moves in the first direction or disposed in the first housing as the second housing moves in the second direction, and a support plate (e.g., a support plate **440** of FIG. **4A**) configured to support the display. According to an embodiment, the support plate includes rigid regions (e.g., rigid regions **440a'** and **440a''** of FIG. **4A**), a flexible region (e.g., a first flexible region **440b'** of FIG. **4A**) disposed between the rigid regions and including a plurality of openings spaced apart from each other and having a length in a third direction (e.g., +x direction) perpendicular to the first direction, a first layer (e.g., a first layer **441** of FIG. **4A**) including a plurality of

first openings extending in the third direction in the flexible region, and a second layer (e.g., a second layer **442** of FIG. **4A**) disposed on the first layer and including a plurality of second openings partially overlapping each of the plurality of first openings in the flexible region. According to an embodiment, a distance (e.g., **d3** in FIG. **7B**) from one end (e.g., one end **441a'** of one first opening of FIG. **7A**) to another end (e.g., the other end **441a''** of one first opening of FIG. **7A**) of one first opening (e.g., one first opening **441a** of FIG. **6A**) of the plurality of first openings facing the third direction may be different from a distance (e.g., **d4** of FIG. **7B**) from the one end of the first opening to one end (e.g., the other end **442a''** of one second opening in FIG. **7A**) of both ends of one second opening (e.g., one second opening **442a** of FIG. **6A**) of the plurality of second openings overlapping the first opening, the one end of the second opening being near to the another end of the first opening.

According to an embodiment, a length (e.g., **d3** of FIG. **7B**) of each of the plurality of first openings in the third direction may be substantially equal to a length (e.g., **d5** of FIG. **7B**) of each of the plurality of second openings in the third direction.

According to an embodiment, the first layer may further include a first bridge (e.g., one first bridge **441b** of FIG. **6A**) contacting the one end of the first opening, and the second layer may further include a second bridge (e.g., one second bridge **442b** of FIG. **6A**) contacting the one end of the second opening. According to an embodiment, a distance (e.g., **s1** in FIG. **7D**) from the another end of the first opening to the first bridge may be different from a distance (e.g., **s2** in FIG. **7D**) from another end of the one second opening (e.g., another end **442a''** of one second opening) to the second bridge.

According to an embodiment, a distance (e.g., **d3** in FIG. **7B**) from the one end of the one second opening to the first bridge is substantially equal to a distance (e.g., **d5** in FIG. **7B**) from the another end of the one second opening to the first bridge.

According to an embodiment, a thickness (e.g., **d7** in FIG. **10**) of the flexible region is less than or equal to a thickness (e.g., **d8** in FIG. **10**) of the rigid regions.

According to an embodiment, the electronic device may further include an adhesive member (e.g., an adhesive member **640** of FIG. **6A**) disposed between the first layer and the second layer in a region corresponding to the rigid regions among the flexible region and the rigid regions.

According to an embodiment, the support plate further includes a continuous surface (e.g., a planar plate **500** of FIG. **5B**) in the planar region. According to an embodiment, the rigid regions and the flexible region disposed in a region correspond to the deformable region.

According to an embodiment, the electronic device may further include at least one support bar (e.g., at least one support bar **1100** of FIG. **11A**) supporting the support plate. According to an embodiment, when viewing the display from above, the at least one support bar may overlap each of the rigid regions.

According to an embodiment, the electronic device may further include at least one gear head (e.g., a gear head **620** of FIG. **6A**) disposed on both peripheries of the support plate extending in a direction parallel to the first direction in the rigid regions), and a guide gear (e.g., a guide gear **630** of FIG. **6A**) engaged with the at least one gear head and guiding movement of the display.

According to an embodiment, the support plate may further include a first reinforcement layer (e.g., a first

reinforcement layer **1110** of FIG. **11A**) disposed between the first layer and the second layer.

According to an embodiment, the support plate may further include at least one of carbon fiber reinforced polymer (CFRP), glass fiber reinforced plastics (GFRP), or stainless used steel (SUS).

According to an embodiment, the support plate may further include a third layer (e.g., a third layer **743** of FIG. **7E**) disposed on the second layer and including a plurality of third openings partially overlapping each of the plurality of first openings and partially overlapping each of the plurality of second openings in the flexible region. According to an embodiment, the distance from the one end to the another end of the one first opening of the plurality of first openings facing the third direction may be different from a distance (e.g., **d6** of FIG. **7E**) from the one end of the one first opening to one end of both ends of one third opening (e.g., one third opening **743a** of FIG. **7E**) of the plurality of third openings overlapping the one first opening, the one end (e.g., one end **743c'** of the third opening disposed in a third direction (e.g., +x direction) with respect to one third opening **743a** of FIG. **7E**) of the one third opening being near to the another end of the first opening.

According to an embodiment, the first layer may further include a first bridge contacting the one end of the one first opening, the second layer may further include a second bridge contacting the one end of the one second opening, and the third layer may further include a third bridge (e.g., one third bridge **743b** of FIG. **7E**) contacting the one end of the one third opening. According to an embodiment, a distance (e.g., **s3** in FIG. **7E**) from the first bridge to the second bridge may be substantially equal to a distance (e.g., **s4** in FIG. **7E**) from the second bridge to the third bridge and may be substantially equal to a distance (e.g., **s5** in FIG. **7E**) from the third bridge to the another end of the one first opening facing the one end of the one first opening.

According to an embodiment, the electronic device may further include an adhesive member disposed between the second layer and the third layer in a region corresponding to the rigid regions among the flexible region and the rigid regions.

According to an embodiment, the support plate may further include a second reinforcement layer (e.g., a second reinforcement layer **1120** of FIG. **11B**) disposed between the second layer and the third layer.

According to an embodiment, an electronic device may include a first housing, a second housing coupled to the first housing so as to be slidable with respect to the first housing in a first direction and a second direction opposite to the first direction, a display including a planar region disposed on the second housing and a deformable region that contacts the planar region and is exposed to an outside as the second housing moves in the first direction or disposed in the first housing as the second housing moves in the second direction, a support plate configured to support the display, at least one gear head disposed on both peripheries of the support plate extending in a direction parallel to the first direction, and a guide gear guiding a movement of the display and rotating meshed with the at least one gear. According to an embodiment, the support plate includes rigid regions, a flexible region disposed between the rigid regions and including a plurality of openings spaced apart from each other and having a length in a third direction perpendicular to the first direction, a first layer including a plurality of first openings extending in the third direction and a plurality of first bridges separating each of the plurality of first openings in the flexible region, and a second layer

disposed on the first layer and including a plurality of second openings partially overlapping each of the plurality of first openings and a plurality of second bridges separating each of the plurality of second openings in the flexible region.

According to an embodiment, a distance from one end of one first opening of the plurality of first openings facing the third direction to one first bridge of the plurality of first bridges in contact with another end of the one first opening may be different from a distance from the one end of the one first opening to a second bridge of the plurality of second bridges, the second bridge being near to the another end of the first opening.

According to an embodiment, a length of each of the plurality of first openings in the third direction may be substantially equal to a length of each of the plurality of second openings in the third direction.

According to an embodiment, a distance from one end of one second opening of the plurality of second openings overlapping the one first opening to the first bridge may be substantially equal to a distance from another end of the one second opening to the first bridge.

According to an embodiment, the electronic device may further include at least one support bar supporting the support plate. According to an embodiment, when viewing the display from above, the at least one support bar may overlap each of the rigid regions.

According to an embodiment, the support plate may further include a third layer disposed on the second layer and including a plurality of third openings partially overlapping each of the plurality of first openings and partially overlapping each of the plurality of second openings, in the flexible region. According to an embodiment, the distance from the one end to the another end of the one first opening of the plurality of first openings facing the third direction may be different from a distance from the one end of the one first opening to one end of both ends of one third opening of the plurality of third openings overlapping the one first opening, the one end of the one third opening being near to the another end of the first opening.

According to an embodiment, the third layer may further include a third bridge in contact with the one end of the one third opening. According to an embodiment, a distance from the first bridge to the second bridge may be substantially equal to a distance from the second bridge to the third bridge, and may be substantially equal to a distance from the third bridge to the other end facing one end of the one first opening.

The electronic device according to various embodiments disclosed herein may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B,

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or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may be interchangeably used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment of the disclosure, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

Various embodiments as set forth herein may be implemented as software (e.g., a program **140**) including one or more instructions that are stored in a storage medium (e.g., an internal memory **136** or an external memory **138**) that is readable by a machine (e.g., an electronic device **101**). For example, a processor (e.g., a processor **120** of an electronic device **101**) of the machine may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

According to an embodiment of the disclosure, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., a compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer’s server, a server of the application store, or a relay server.

According to various embodiments of the disclosure, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments of the disclosure, one or more of the

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above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments of the disclosure, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments of the disclosure, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

What is claimed is:

1. An electronic device comprising:

a first housing;

a second housing coupled to the first housing so as to be slidable with respect to the first housing in a first direction and a second direction opposite to the first direction;

a display including a planar region disposed on the second housing and a deformable region that contacts the planar region, wherein the deformable region is exposed to an outside as the second housing moves in the first direction or disposed in the first housing as the second housing moves in the second direction; and

a support plate configured to support the display;

wherein the support plate includes:

rigid regions;

a flexible region disposed between the rigid regions, the flexible region including a plurality of openings spaced apart from each other and having a length in a third direction perpendicular to the first direction;

a first layer including a plurality of first openings extending in the third direction in the flexible region; and

a second layer disposed on the first layer and including a plurality of second openings partially overlapping each of the plurality of first openings in the flexible region;

wherein a distance from one end to another end of one first opening of the plurality of first openings facing the third direction is different from a distance from the one end of the one first opening to one end of both ends of one second opening of the plurality of second openings overlapping the one first opening, the one end of the one second opening being near to the another end of the first opening.

2. The electronic device of claim 1, wherein a length of each of the plurality of first openings in the third direction is substantially equal to a length of each of the plurality of second openings in the third direction.

3. The electronic device of claim 1, wherein the first layer further includes a first bridge contacting the one end of the one first opening,

wherein the second layer further includes a second bridge contacting the one end of the one second opening, and

wherein a distance from the another end of the one first opening to the first bridge is different from a distance from another end of the one second opening to the second bridge.

4. The electronic device of claim 3, wherein a distance from the one end of the one second opening to the first bridge is substantially equal to a distance from the another end of the one second opening to the first bridge.

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5. The electronic device of claim 1, wherein a thickness of the flexible region is less than or equal to a thickness of the rigid regions.

6. The electronic device of claim 1, further comprising an adhesive member disposed between the first layer and the second layer in a region corresponding to the rigid regions among the flexible region and the rigid regions.

7. The electronic device of claim 1, wherein the support plate further includes a continuous surface in the planar region, and

wherein the rigid regions and the flexible region disposed in a region correspond to the deformable region.

8. The electronic device of claim 1, further comprising: at least one support bar supporting the support plate, wherein, when viewing the display from above, the at least one support bar overlaps each of the rigid regions.

9. The electronic device of claim 1, further comprising: at least one gear head disposed on both peripheries of the support plate extending in a direction parallel to the first direction in the rigid regions; and

a guide gear meshed with the at least one gear head and guiding a movement of the display.

10. The electronic device of claim 1, wherein the support plate further includes a first reinforcement layer disposed between the first layer and the second layer.

11. The electronic device of claim 1, wherein the support plate further includes at least one of carbon fiber reinforced polymer (CFRP), glass fiber reinforced plastics (GFRP), or stainless used steel (SUS).

12. The electronic device of claim 1, wherein the support plate further includes a third layer disposed on the second layer and including a plurality of third openings partially overlapping each of the plurality of first openings and partially overlapping each of the plurality of second openings in the flexible region, and

wherein the distance from the one end to the another end of the one first opening of the plurality of first openings facing the third direction is different from a distance from the one end of the one first opening to one end of both ends of one third opening of the plurality of third openings overlapping the one first opening, the one end of the one third opening being near to the another end of the first opening.

13. The electronic device of claim 12, wherein the first layer further includes a first bridge contacting the one end of the one first opening,

the second layer further includes a second bridge contacting the one end of the one second opening,

the third layer further includes a third bridge contacting the one end of the one third opening, and

a distance from the first bridge to the second bridge is substantially equal to a distance from the second bridge to the third bridge and is substantially equal to a distance from the third bridge to the another end of the one first opening facing the one end of the one first opening.

14. The electronic device of claim 12, further comprising an adhesive member disposed between the second layer and the third layer in a region corresponding to the rigid regions among the flexible region and the rigid regions.

15. The electronic device of claim 12, wherein the support plate further includes a second reinforcement layer disposed between the second layer and the third layer.

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16. An electronic device comprising:

a first housing;

a second housing coupled to the first housing so as to be slidable with respect to the first housing in a first direction and a second direction opposite to the first direction;

a display including a planar region disposed on the second housing and a deformable region that contacts the planar region, wherein the deformable region is exposed to an outside as the second housing moves in the first direction and is disposed in the first housing as the second housing moves in the second direction;

a support plate configured to support the display;

at least one gear head disposed on both peripheries of the support plate extending in a direction parallel to the first direction; and

a guide gear guiding a movement of the display and rotating meshed with the at least one gear;

wherein the support plate includes:

rigid regions;

a flexible region disposed between the rigid regions, the flexible region including a plurality of openings spaced apart from each other and having a length in a third direction perpendicular to the first direction; a first layer including a plurality of first openings extending in the third direction and a plurality of first bridges separating each of the plurality of first openings in the flexible region; and

a second layer disposed on the first layer and including a plurality of second openings partially overlapping each of the plurality of first openings and a plurality of second bridges separating each of the plurality of second openings in the flexible region; and

wherein a distance from one end of one first opening of the plurality of first openings facing the third direction to one first bridge of the plurality of first bridges in contact with another end of the one first opening is different from a distance from the one end of the one first opening to a second bridge of the plurality of second bridges, the second bridge being near to the another end of the first opening.

17. The electronic device of claim 16, wherein a length of each of the plurality of first openings in the third direction is substantially equal to a length of each of the plurality of second openings in the third direction.

18. The electronic device of claim 16, wherein a distance from one end of one second opening of the plurality of second openings overlapping the one first opening to the first bridge is substantially equal to a distance from another end of the one second opening to the first bridge.

19. The electronic device of claim 16, further comprising: at least one support bar supporting the support plate, wherein, when viewing the display from above, the at least one support bar overlaps each of the rigid regions.

20. The electronic device of claim 16, wherein the support plate further includes a third layer disposed on the second layer and including a plurality of third openings partially overlapping each of the plurality of first openings and partially overlapping each of the plurality of second openings, in the flexible region, and

wherein the distance from the one end to the another end of the one first opening of the plurality of first openings facing the third direction is different from a distance from the one end of the one first opening to one end of both ends of one third opening of the plurality of third

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openings overlapping the one first opening, the one end of the one third opening being near to the another end of the first opening.

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